Beneath the Skin: Interactive Biomechanical Facial Simulations via Composable Co-Expressions

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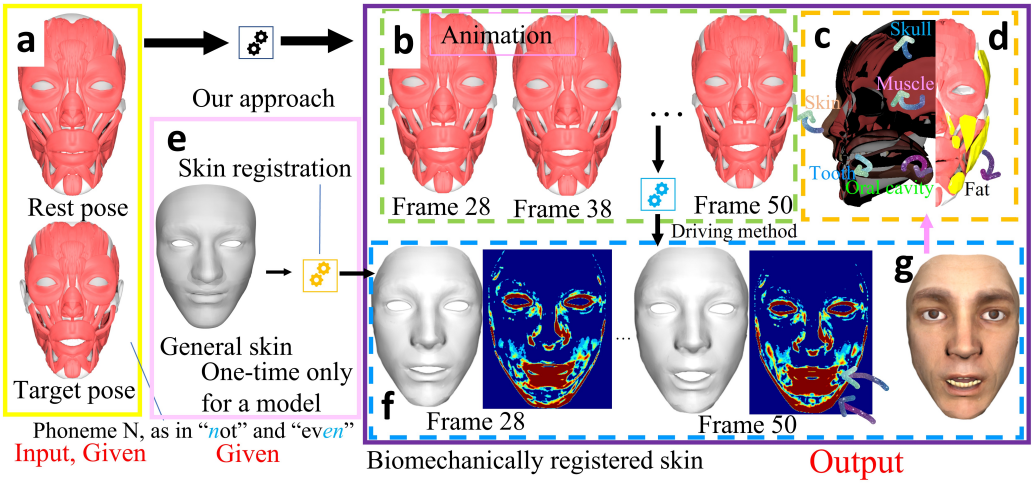

Fig. 1. Overview of our physically and biomechanically based facial animation framework. Our approach generates visually realistic and anatomically accurate expressions by propagating internal muscle dynamics directly to an epidermal skin layer. a. The system input consists of an arbitrary rest pose and an artist-sculpted target pose. b. By jointly minimizing Euclidean distance and biomechanical energy consumption, we compute the continuous deformation trajectory of the internal musculature. c. A sagittal cross-section revealing the multi-layered internal anatomy. d. A topologically reduced musculature proxy mesh, augmented with fat compartments, utilized for skin registration. e. A generic template skin is physically registered to the personalized internal tissues. This physical registration is a one-time, offline preprocessing step per character model. f. The dynamic sequence of the registered skin mesh, faithfully driven by the force propagation from the underlying active muscle layers, with the heatmap of the pixel change compared to the rest pose. g. The final textured and rendered skin at the target pose; the red dashed box indicates the region of the cross-section shown in Fig. 1d.

Realistic facial animation plays a pivotal role in animating convincing, expressive characters. While kinematic techniques offer great artistic engagement, and motion capture offers explicit human performances, biomechanical simulation promises physical plausibility that respects volume preservation, passive dynamics, momentum, deformation, and gravity. However, physical approaches have often come at the price of reduced artistic engagement due to the mismatch between the complexity and constraints of biomechanical motion synthesis and the expressive flexibility demanded by artists. We argue that rigorous biomechanical physics and intuitive artistic control are not mutually exclusive; they can be partially bridged through the generation of physical motor controllers that interpolate facial configurations and action units, including speech. We chose the FACS model [Ekman and Friesen 1978] that is commonly used by character animators, although motions, i.e., muscle activation sequences, can be synthesized for any set of physically plausible facial poses. These controllers may then be composed into complex, realistic motion sequences. Furthermore, the optimal simulation of these sequences requires muscle activation co-expression in a manner that is consistent with coarticulation in speech synthesis. While our primary goal is to demonstrate "under the skin" motion synthesis, we describe a skinning technique that attaches to our musculoskeletal model to demonstrate the potential of fully biomechanical, expressive, artist-directable facial animation.

CCS Concepts: • **Computing methodologies** → **Simulation by animation**.

Additional Key Words and Phrases: Facial animation, Physically based methods, biomechanically based methods, gradient descent

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1 Introduction

The demand for high-fidelity character animation is growing rapidly, driven by the proliferation of advanced game engines and character creation platforms [Xie et al. 2024]. Within this broad domain, facial animation remains a critical focus due to the inherent complexity and fine-grained nuances of human expressions. The applications

for robust facial animation span several diverse fields. In the entertainment sector, it is foundational for film production, interactive gaming, and the creation of digital twins. Furthermore, it plays an increasingly vital role in modern virtual conferencing technologies (such as Apple Persona or Meta's Codec Avatars), where authentic facial reconstruction is necessary to digitally replace faces occluded by virtual reality headsets. Beyond entertainment and communication, physically grounded facial animation holds significant potential in medicine and biomechanics. Accurate simulations can facilitate surgical planning and physical therapy, enhance medical education by visually exposing internal facial anatomy [Zheng and Barbic 2024], and allow scientists to study human facial mechanics to directly inform the design of humanoid robots [Zheng et al. 2022].

However, modern facial animation faces four primary challenges. The first is achieving visual realism, as the human eye is acutely sensitive to even minor unnatural surface deformations. The second is ensuring physical and biomechanical accuracy; applications in both medicine and high-fidelity visual effects demand that tissue dynamics are strictly grounded in real-world physics rather than mere visual plausibility. The third challenge relates to labor efficiency. Traditional film and game production pipelines require a massive investment of time from skilled artists to manually sculpt and refine complex expressions, creating a critical need for automated methods that drastically reduce manual labor while preserving quality. Finally, there is a fundamental tension between computational speed and interactive application demands. Rigorous biomechanical simulations are traditionally computationally expensive and slow, yet emerging applications like virtual conferencing require extremely fast, real-time simulation speeds to be viable.

1.1 Problem statement

Our goal is to develop an approach that generates physically plausible facial animations under stringent constraints by leveraging physical simulation [Zheng et al. 2022]. At the foundational level, our computational framework takes as input the vertices of distinct musculoskeletal meshes—representing individual muscles and internal tissues—in their respective rest poses, alongside the vertices of the corresponding prescribed target poses. Rather than solving a single monolithic system, our framework treats these as individual, specialized subsystems simulated in a single pass, utilizing a synchronization mechanism to coordinate their physical collaboration. Instead of simply interpolating between these states, the system solves a coupled inverse dynamics-like problem: it simultaneously calculates physically correct deformation trajectories across all submeshes and outputs a set of synchronized physical motor controllers. Crucially, this optimization relies on the assumption that tracking a sparse set of salient structural landmarks is sufficient to reconstruct the continuous deformation of the entire volume. We assume that the spatial distribution of these landmarks captures enough geometric variance to accurately map and drive all remaining vertices across the high-resolution meshes. From the user's perspective, however, this mathematical complexity is entirely abstracted. An animator simply provides high-level inputs—ranging from a sequence of discrete activations via a piano-style interface to natural language prompts (e.g., "not even you")—and the system outputs a fluid, biomechanically accurate facial animation that automatically includes the synchronous deformation of muscles, internal tissues, and skin.

Currently, the animation industry relies heavily on mature, traditional kinematic methods, predominantly blendshapes (linear interpolation between predefined poses) [Anjyo 2018], to achieve visual realism [Bao et al. 2019]. While these ubiquitous tools offer direct artistic control, they are inherently disconnected from human biomechanics. This lack of physical constraints limits their utility not only in physically grounded visual effects but also in emerging applications such as physiology, facial surgery planning, and medical therapy. To bridge this gap, accurate models incorporating inherent physical attributes—such as gravity, tissue volume preservation, and collision constraints—are required [Zheng et al. 2022]. However, historically, muscle-based models and full finite element methods (FEM) [Ichim et al. 2017] have been computationally prohibitive, especially for high-resolution 3D assets. Furthermore, artists are understandably reluctant to abandon their efficient, well-established kinematic pipelines unless a physically based method offers both a compelling visual upgrade and an accessible, familiar workflow.

While blendshapes excel at reaching specific, predefined targets, their kinematic nature presents several drawbacks. First, combining complex or contradictory expressions often yields unnatural geometric intersections. Second, the linear transition between discrete poses frequently introduces volume-loss artifacts, such as the typical "shrink" effect observed during the middle frames of kinematic interpolation [Ma et al. 2012]. Finally, while both physical and kinematic frameworks rely on artist-defined target poses, traditional blendshape pipelines require a massive, manual investment of time to sculpt an exhaustive library of individual expressions and discrete shape correctives to handle combinations. In contrast, our framework requires only a minimal set of target states; because our physical motor controllers are inherently composable, the system automatically and dynamically computes complex, multi-expression combinations from that baseline, drastically reducing the artist's manual workload.

1.2 Argument

To address these limitations without sacrificing usability, we argue that rigorous biomechanical physics and intuitive artistic control are not mutually exclusive. Rather, they can be successfully bridged through the generation of physical motor controllers that interpolate facial configurations and action units, including complex motions such as speech. To realize this, we propose a biomechanically based framework that dynamically generates these animations through physical optimization. As illustrated in Figure 2, an artist prescribes a high-level target pose; our system then samples representative landmarks across the tissue and employs a gradient descent solver to actuate the underlying musculoskeletal structure toward that target.

Our facial model is capable of deforming to a diverse array of expressions and is able to realize lip synchronization driven by natural language commands. While the Facial Action Coding System (FACS) provides the foundational basis for these expressions, our model explicitly computes the continuous spatial distribution of

tissue deformation. This continuous formulation allows the mesh to converge on any arbitrary physical pose, enabling the synthesis of highly nuanced micro-expressions well beyond the standard FACS vocabulary. This underlying optimization approach has two primary goals: (1) to minimize energy consumption within human biomechanical constraints, ensuring physical accuracy; and (2) to minimize geometric Euclidean distance, ensuring visual realism."

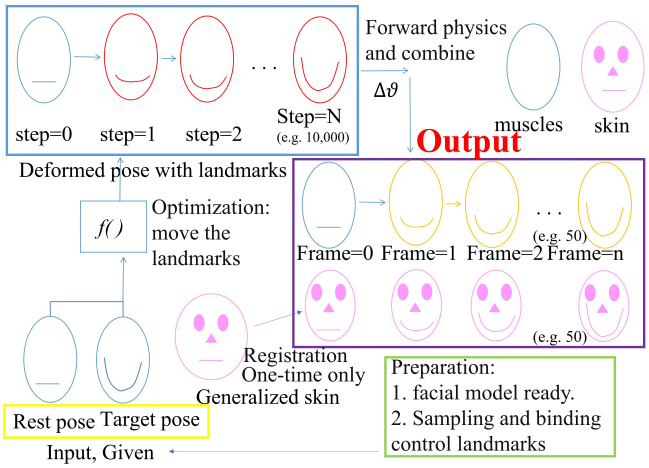

Fig. 2. The pipeline of our approach.

1.3 Contribution

The potential contributions and innovations are as follows:

- We introduce a physically and biomechanically grounded facial animation framework that explicitly models and animates distinct internal tissue structures. This modular approach facilitates granular biomechanical analysis, broadening its potential utility for medical applications. To mitigate the computational overhead of dense vertex manipulation, we propose a sparse landmark sampling scheme that efficiently proxies the underlying high-resolution tissue meshes. Finally, we actuate these landmarks via an optimization-based solver, granting rigorous and precise control over the resulting complex tissue dynamics.
- To emulate the inherent energy efficiency of human motor control, we incorporate a comprehensive energy-minimization objective directly into our optimization framework. Specifically, our gradient formulation integrates active actuation—driven by a Hill-type muscle model—with passive elastic and strain energies. This physiologically grounded design ensures physically plausible deformations while providing a robust foundation for analyzing complex biomechanical phenomena, such as phonetic coarticulation and subtle micro-expressions.
- We introduce a highly composable expression transition framework that bridges interactive artistic control with underlying biomechanics. By pre-computing deformation parameters into a transition table, the system empowers artists to intuitively sequence target expressions and dictate their temporal overlap. This overlapping activation naturally yields smooth, realistic co-expression while maintaining biological energy efficiency. To robustly execute this at the tissue level, we propose a novel capping algorithm and a servo-based optimizer, which precisely synchronize the concurrent physical optimizations across multiple interacting anatomical layers.
- To biomechanically conform the epidermal layer to the internal anatomy, we formulate a cloth-based simulation framework to drape a generic, high-resolution skin mesh over subject-specific skeletal structures, muscles, and adipose tissues. Furthermore, we dynamically couple this skin layer to the underlying anatomy using a network of elastic spring attachments and wire deformers. This ensures that the biomechanically accurate volumetric muscle deformations naturally and realistically drive the surface geometry, completing our comprehensive, "inside-out" facial modeling pipeline.

1.4 Organization of the Paper

The remainder of this paper is organized as follows: Section 2 reviews the related work. Section 3 details the computational architecture of our physical motor controllers, demonstrating how they abstract complex musculoskeletal dynamics to provide intuitive, kinematic-style artistic control. Section 4 presents our experimental results and quantitative analyses, validating that our composable framework successfully bridges rigorous biomechanical physics with expressive animation constraints. Section 5 provides a comprehensive discussion of our methodology, contrasting our automated, physics-driven approach with existing kinematic and fully finite-element alternatives. It also outlines future avenues for research in biomechanical motion synthesis. Finally, Section 6 concludes the manuscript by summarizing our core contributions.

2 Related work

The landscape of facial animation has historically been divided into two distinct paradigms: kinematically based methods, which prioritize expressive artistic control, and physically based methods, which prioritize anatomical and biomechanical accuracy. More recently, data-driven approaches have emerged as a popular alternative to balance visual realism with computational efficiency. However, as explored in this section, existing methods across all three categories consistently force a trade-off between intuitive usability and rigorous physical grounding. In the following, we review these domains to highlight this persistent historical separation, thereby establishing the precise gap in the literature that our proposed framework—which bridges biomechanical physics and intuitive artistic control—is designed to resolve.

Nowadays, the most commonly used methods in films are kinematically based methods, such as the blend shapes method. The blend shapes method connects a rest shape and some target shapes by interpolating the interval vertex positions of the mesh, and thus produces an animation. While kinematic methods have the merits of fast computing, they require large labor workloads from artists and often lack physical realism. In academia, kinematic methods are usually combined with other animation methods. Close to our work, Bao et al. [2019] proposed a motion capture-driven facial

model. Using blend shapes method, the rest pose deforms to the camera-captured target pose, and this deformation can drive the muscle model to activate and move. However, it needs actors to drive the landmarks and the face model, which has limited deformation freedom and requires expensive actor resources. In comparison, in our work, we do not use motion capture from human actors to drive landmarks. Instead, an optimization-based approach will drive the landmarks towards the target pose.

Another theme is to employ physically based methods, which aims to simulate the underlying biomechanics of facial tissues, such as muscles, skin, and bones, to produce realistic animations. These methods often achieve higher realism but can be computationally expensive and require detailed anatomical models. Zheng et al. [2022] developed a precise anatomically based hand model using magnetic resonance imaging. They achieve this by combining elastoplastic simulation with magnetic resonance imaging (MRI) scans. Unlike Zheng et al. [2022], we do not adopt the fully physical hand model to our facial model, because we find the onerous non-academic work behind this paper unaffordable: making many rubber cases for the target poses, to stabilize the hand for minutes under MRI scans. Also, the computational time for the hand model takes hours to converge, whereas by introducing representative landmarks, we obtain a better trade-off for accuracy and efficiency.

Furthermore, Ichim et al. [2017] proposed a full finite element method for facial animation. Firstly, they solve the inverse physics to obtain the activation parameters for muscles. Secondly, they solve the forward physics to deform the rest pose to the target pose by finite element methods, with the goal of minimizing the energy. Unlike [Ichim et al. 2017], we do not directly control the large number of vertices of the face model. Instead, we sample landmarks to represent the mesh of a tissue, which is more efficient in computation. Also, we use an optimization-based approach to drive the landmarks, which offers us more control over the animation. Additionally, Ichim et al. [2017] models the face as a whole volume with spatially varying plastic strains [Zheng et al. 2022], whereas we model the tissues in the face separately and animate internal facial anatomy explicitly, which enables finer biomechanical interpretation and potential application in medicine.

Similarly addressing real-time performance, Fernández-Fernández et al. [2025] propose an interactive facial animation framework that pre-computes a global material stiffness map across a set of target expressions. At runtime, the system calculates the equilibrium of potential energy to continuously track the driving forces of a kinematic rig. However, because their simulation is restricted to a superficial, surface-only shell, it fundamentally lacks internal anatomical structure. In contrast, our approach explicitly simulates the underlying volumetric musculoskeletal dynamics, anchoring the skin's surface deformation to a highly accurate physical foundation. Moreover, Corigliano et al. [2025] introduced a self-supervised neural animation model that, during training, evaluates geometric predictions against physical energy minimization and collision penalties. At runtime, the model bypasses traditional simulation entirely, utilizing the trained neural network to infer the final equilibrium states of the skin, bone, and teeth. While highly performant, this relies on a black-box neural approximation of physics. Our approach, conversely, eschews neural inference; we explicitly simulate the true, causal mechanical chain of human expression from the inside out. By directly computing the physical laws governing tension and compression, we preserve the mathematical transparency required for rigorous biomechanical and medical studies.

Data-driven methods use datasets as the knowledge to train a neural network and infer the deformed face mesh. These methods are highly efficient at runtime and are able to produce realistic results, but they shrink the deformation space of the mesh to the existing knowledge in datasets, limiting the expressiveness of the model. Also, since it lacks volume preservation, artifacts usually appear when there is pressure on the surface of a tissue, such as pressure between the lips and between the eyelids. Furthermore, they often require extensive training data and may lack generalization to unseen, out-of-distribution expressions. Park et al. [2024] developed a near-realtime facial animation method using neural networks. Firstly, they employ a finite element method to generate a dataset that contains high-resolution meshes, and use downsampling to obtain the corresponding low-resolution facial pose meshes. Moreover, the high and low-resolution pairs can train a neural network to learn the mapping from coarse mesh to precise mesh. At the testing set, the trained neural network can output a sequence of precise meshes to produce an animation given a sequence of low-resolution meshes.

Chu et al. [2025] proposes to use neural networks to approximate the physically based facial animation. Their inputs are the speech voice, and their outputs are the animation of the speech. Firstly, Chu et al. [2025] employs an encoder to convert the speech voice to a sequence of latent features, then the approach maps the latent features to the activation setup of facial muscles and produces the animation. Nicolas et al. [2024] also proposes a neural network-based method to approximate physically based facial animations. They use finite element methods to calculate deformation parameters, and then use a neural network to learn to infer these deformation parameters. Unlike data-driven methods, our method adopts the artist-prescribed pose as the final pose, and we employ a gradient descent method to search for trajectories towards the final poses, and there is no dataset or neural networks involved. This allows us to avoid the limitations of data-driven methods, such as reduced expressiveness and artifacts, while maintaining full control over the animation process.

Existing registration methods heavily rely on external observational data. For instance, several approaches use non-rigid Iterative Closest Point (ICP) techniques to map template meshes to 3D facial scans [Ichim et al. 2017; Yang et al. 2024]. Similarly, recent volumetric models register skin and internal tissues directly to MRI data [Li et al. 2022; Zheng et al. 2022], while inverse-physics frameworks extract parameters from video to fit cloth to moving bodies [Zheng et al. 2024].

In contrast, our approach entirely bypasses the need for external scans or observational data. Instead of relying on surface-level captures, we generate the epidermal topology strictly from the inside out by physically conforming the skin mesh directly to the volumetric landscape of the underlying musculature. This scan-free formulation represents a significant advantage for digital character creation, as it natively supports purely fictitious or stylized characters that lack real-world counterparts.

3 Approach

In this section, we detail the computational framework designed to bridge rigorous biomechanical physics with intuitive artistic control. Rather than relying on purely kinematic deformation, we construct a volumetric musculoskeletal facial model and employ an optimization-based solver to generate composable physical motor controllers. By driving a sparse set of structural landmarks toward artist-defined targets, our system automatically synthesizes physically plausible, continuous tissue deformations that satisfy both biomechanical constraints and strict demands for visual realism.

3.1 Formulation

To automatically generate the physical motor controllers capable of driving complex expressions, we must first establish a mathematical representation of the underlying anatomy. Because our solver computes the physical forces required to transition the face from a rest state to a target expression, we begin by parameterizing the system's primary actuators: the muscular layer. Assume we have n muscles within a specific anatomical regionof the face model, denoted as $m_1, m_2, ..., m_n$. Furthermore, assume we have p landmarks in the face model. Landmarks are the virtual joints in the rigging (virtual bones control) hierarchy, such as the green or yellow spheres in Figure 3a and Figure 3b. The landmarks in Figure 3b serve to rotate the mandible. The landmarks in Figure 3a (the connecting virtual bones to their parent virtual joints are hidden for conciseness), also referred to as "rigged joints" in other papers, are representative vertices selected from the vertices of a tissue to represent the whole tissue mesh. At the position of a selected vertex, we assign a virtual landmark, and bind it to the surrounding 20 vertices based on the geometric adjacency. When one landmark moves, it pulls the surrounding vertices to move together in the same direction. In Figure 3c, the red area depicts the affected vertices controlled by one landmark in the middle of the red area.

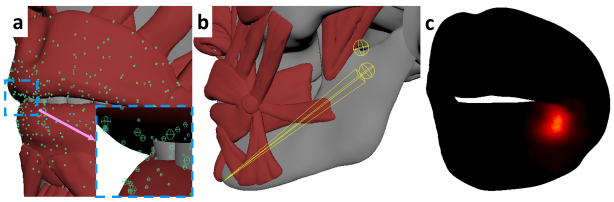

Fig. 3. a. The 300 sample landmarks distribution. b. The simulation of bones. c. The illustration of a landmark and the affected vertices.

We use Euclidean angle to represent the rotation differential of each landmark: $\mathbf{d}_1, \mathbf{d}_2, ..., \mathbf{d}_p$, where $\mathbf{d} \in \mathbb{R}^3$. The translation values of the vertex of the landmarks are $\mathbf{b}_1, \mathbf{b}_2, ..., \mathbf{b}_p$, where $\mathbf{b} \in \mathbb{R}^3$. We want the distance sum between the ground truth vertices (prescribed by the artist) and the vertices calculated by our approach to be small.

Let $\mathbf{x_i} \in \mathbb{R}^4$ be the ground truth position of the i-th vertex dictated by the artist; note that the fourth dimension of $\mathbf{x_i}$ is the padded 1 to form the homogeneous four-tuple representation of a 3D point, for the generalization of the equation in translation. Let $\bar{\mathbf{x}}_\mathbf{i} \in \mathbb{R}^4$ be the rest pose position of the i-th vertex. Let $\mathbf{y_i} \in \mathbb{R}^4$ be the i-th vertex position produced by our approach.

Although $\mathbf{x_i}$ and $\mathbf{y_i}$ are given, there are several reasons why we do not assign the value of $\mathbf{x_i}$, or the interpolated value of $\mathbf{x_i}$, directly to $\mathbf{y_i}$ (i.e., the simplest kinematically based method): 1) we do not directly control the movement of vertices, but only through the control of landmarks and bones. In fact, taking control of all the vertices in a high-resolution object can lead to a dimension blow-up. Also, in a complex facial model 3D mesh, directly moving the vertices or faces of the mesh will make Maya crash immediately, because this operation inevitably incurs extremely small angles in the triangle mesh of the face model, which exceeds the capability of most physical calculation simulators. And, 2) we want to achieve a more physically accurate animation process than the blendshapes method does.

Then we stack the vertices' position vector into the matrix:

$$\bar{\mathbf{x}} = [\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2, ..., \bar{\mathbf{x}}_\mathbf{N}], \mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_\mathbf{N}], \mathbf{y} = [\mathbf{y}_1, \mathbf{y}_2, ..., \mathbf{y}_\mathbf{N}], \quad (1)$$

To find $\mathbf{y}$, we need to find the deformation differential variables, where we are given two inputs: the rest pose position of the vertices and the target pose position of the vertices.

$$\Delta\boldsymbol{\theta} = f(\bar{\mathbf{x}}, \mathbf{x}), \quad (2)$$

where $\Delta\boldsymbol{\theta}$ consists of two components:

$$\Delta\boldsymbol{\theta} \triangleq [\mathbf{D}, \mathbf{B}], \quad (3)$$

$$\mathbf{D} = [\mathbf{d}_1, ..., \mathbf{d}_p], \mathbf{B} = [\mathbf{b}_1, ..., \mathbf{b}_p], \quad (4)$$

where $\Delta\boldsymbol{\theta} \in \mathbb{R}^{(3p+3p)\times 1}$ is the flattened Maya parameter vectors; by unflattening it, we will obtain the parameters $\mathbf{D}$ (landmarks rotation matrix), and $\mathbf{B}$ (landmarks translation matrix). $f(\cdot)$ is the core control algorithm that deforms one pose to another by finding a combination of control variables (in our case, it is the Maya parameter vectors).

Then we apply the $\Delta\boldsymbol{\theta}$ to rest pose $\bar{\mathbf{x}}$, to obtain the predicted deformed pose position of vertices:

$$\mathbf{y} = h(\bar{\mathbf{x}}, \Delta\boldsymbol{\theta}), \quad (5)$$

where $h()$ is the mapping function; after we apply the parameter change $\Delta\boldsymbol{\theta}$ in Maya software, the Maya simulator will produce the predicted pose position of vertices $\mathbf{y}$. The Maya simulator receives the command $\Delta\boldsymbol{\theta}$, and it performs the translation or rotation of the landmarks. Meanwhile, it calculates how the vertices that are affected by the movement of the landmarks translate along with the movement of the landmarks. At last, it outputs the final pose $\mathbf{y}$.

Note that by changing parameters $\mathbf{d}_i$ or $\mathbf{b}_i$, it is the whole translation matrix $\mathbf{M}$ that is changed, not merely the single $\mathbf{M_i}$, because parameters $\mathbf{d}_i$ or $\mathbf{b}_i$ do not merely influence a single vertex, but at least a region of vertices.

Aside from the position of the vertices, we also use the position of landmarks and Euclidean angles of landmarks with regard to the coordinate axes.

3.2 Objective Function

Our objective function balances three physically motivated terms—surface distance, elastic strain energy, and muscle work—each normalized to a comparable scale:

$$J = w_1 \underbrace{\frac{d_{\text{remain}}}{d_{\text{init}}}}_{\text{normalized Distance}} + w_2 \underbrace{\frac{\frac{1}{2}\,k\,(\Delta L \cdot s)^2}{E_{\max}}}_{\text{normalized Strain Energy}} + w_3 \underbrace{\frac{c \cdot s}{W_{\max}} \sum_i F_{\text{muscle},i}\,\Delta p_i}_{\text{normalized Muscle Work}} \tag{6}$$

where the remaining distance d_{remain} between the current vertex positions $\mathbf{x}$ and the target positions $\mathbf{y}$ is defined as MTAO: The use of numerical derivatives here is fairly problematic for performance, now that we're not limited by the deadline I think a major investment should go into improving the performance. I am happy with the pathway of using numerical derivatives to get a prototype → building up an efficient version. □

$$d_{\text{remain}} = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{\sum_i \sum_j (x_{ij} - y_{ij})^2}, \tag{7}$$

d_{init} is the initial distance $\|\mathbf{x}_0 - \mathbf{y}\|_2$ measured between the rest-pose vertices $\mathbf{x}_0$ and the target vertices $\mathbf{y}$ at the beginning of each optimisation phase, serving as the normalization constant so that the distance term starts at 1 and decreases toward 0; w_1, w_2, w_3 are scalar weights for the distance, strain-energy and muscle-work terms respectively (with default values $w_1 = 0.34$, $w_2 = 0.33$, $w_3 = 0.33$); k is the stiffness of the soft tissue modelled as a linear spring constant (N/m); ΔL is the strain distance, i.e. the Euclidean distance between the current vertex position and the target position in Maya units (under the assumption in Appendix F, in this paper, $\Delta L = d_{\text{remain}}$); s is the unit-conversion scaling factor from Maya units to metres ($s = 9.62 \times 10^{-4}$ m per Maya unit); $E_{\max}$ is the maximum expected strain energy (J) used to normalize the strain-energy term; F_{muscle} is the muscle force acting on the i-th landmark, computed via the Hill muscle model (see Eq. 9); $\Delta p_i = 10^{-3}$ is the translational displacement of the i-th landmark from its rest position. It is the small value that we nudge the mesh and calculate the gradient. c is an amplification factor that scales the muscle-work contribution so that its gradient is not negligibly small; and $W_{\max}$ is the maximum expected muscle work (J) for normalization.

The final reward signal sent to calculate the numerical gradient for the i-th landmark is

$$r_i = -\,d_{\text{init}} \cdot J_i, \tag{8}$$

where the negation converts the minimization objective into a maximisation signal, and the multiplication by d_{init} re-scales the dimensionless objective back to the physical domain.

3.3 Muscle Force Model

Because our optimization solver calculates a physically correct deformation trajectory, it fundamentally requires a robust mathematical definition of the internal forces driving the musculoskeletal system. To accurately simulate this biological actuation, the muscle force F_{muscle} is computed using Hill's muscle model [Hill 1938; Zajac 1989], which decomposes the total force into an active component and passive elastic components:

$$F_{\text{muscle}} = \min\Big(\alpha \cdot f_L(l) \cdot f_V(v) + F_{\text{PEE}} + F_{\text{SEE}},\ F_{\max}\Big), \tag{9}$$

where $\alpha \in [0, 1]$ is the degree of muscle activation, f_L is the force–length relationship, f_V is the force–velocity relationship, F_{PEE} is the passive parallel elastic element force, F_{SEE} is the passive series elastic element force, and $F_{\max}$ is the maximum isometric force. The total force is clamped to not exceed $F_{\max}$.

Force–Length Relationship. The active force–length relationship is modelled as a Gaussian curve centred at the optimal fibre length l_0:

$$f_L(l) = F_{\max} \cdot \exp\left(-\frac{(l - l_0)^2}{2\,\sigma^2\,l_0^2}\right), \tag{10}$$

where $l = l_0 + \bar{\Delta} d$ is the current muscle fibre length, l_0 is the muscle fibre's original (anatomical) length, $\bar{\Delta} d$ is the average remaining distance to the target. Note that in the source [Zajac 1989], $\bar{\Delta} d$ is the average translation of the fibre length. We adapt the theory to our approach by making $\bar{\Delta} d$ the average remaining distance, and the justification can be seen in Appendix F. Moreover, σ controls the width of the length–tension curve. This Gaussian form provides a smooth approximation of the classic sarcomere length–tension relationship [Zajac 1989].

Force–Velocity Relationship. The force–velocity relationship follows the classic Hill equation [Hill 1938]. For concentric contraction ($v \geq 0$):

$$f_V(v) = \frac{(F_{\max} + a)\,b}{v + b} - a, \tag{11}$$

where v is the average landmark velocity (in physical units), a and b are Hill's characteristic constants that shape the hyperbolic force–velocity curve, related to the muscle's maximum shortening velocity $v_{\max}$ and a shape factor K via $b = v_{\max}\,a\,/\,(F_{\max}\,K)$. Note that due to the adaptation in eq 10, we only need to use the concentric contraction (not the eccentric contraction): the muscle length shortens towards the target pose.

Passive Forces. The passive parallel elastic element (PEE) generates force when the muscle is stretched beyond its optimal length:

$$F_{\text{PEE}} = k_{\text{PEE}} \cdot \max(l - l_0,\ 0), \tag{12}$$

where k_{PEE} is the reduced passive stiffness. And the series elastic element (SEE) transmits force through the tendon:

$$F_{\text{SEE}} = k_{\text{SEE}} \cdot l_{\text{tendon}}, \tag{13}$$

where k_{SEE} is the stiffness constants (N/cm) of the series elastic elements, and l_{tendon} is the tendon stretch (cm).

3.4 Optimization

In Section 3.1, the process is to obtain the predicted position $\mathbf{y}$ at time $t+1$ from the rest position $\mathbf{x}$ at time t. However, this small time step can take many epochs for the optimization method to achieve satisfactory results in Eq. 6. Each epoch is an action of a landmark or several landmarks to get closer to the target pose. This section extends the process in Eq. 6.

Intuitively, we calculate the distance between the current pose and the target pose, and use the negative distance as the reward r_d to guide the gradient to update deformation parameters.

Taking orbicularis oris (lip) as an example, let $d(e)$ be the Euclidean distance between the ground truth vertex coordinates of the lip $\mathbf{x}$ and the vertex coordinates of the lip produced by our approach $\mathbf{y}(e)$. e is the current epoch. The converging condition is that the distance $d(e)$ is smaller than its corresponding threshold. This distance threshold is represented by a percentage of the original distance (for example, 1% of the original distance). Furthermore, the constraints are:

$$\mathbf{yj}(e+1) = \mathbf{yj}(e) + \mathbf{B}(e) \text{ or } \mathbf{D}(e), \tag{14}$$

where $\mathbf{yj}(e) \in \mathbb{R}^{n\times 3}$ is the current rotation position of the landmarks of the tissue at epoch e, and $\mathbf{j}$ in this valuable means these landmarks are the virtual rotational joint type of the landmarks. $\mathbf{B}(e) \in \mathbb{R}^{n\times 3}$ is the control effort at epoch e, i.e., the translation of the landmarks, and it is a differential variable. Similarly, $\mathbf{D}(e) \in \mathbb{R}^{n\times 3}$ is the rotational control effort at epoch e. If the landmarks of the tissue move by translation, the equation uses $\mathbf{B}(e)$, otherwise (move by rotation), the equation uses $\mathbf{D}(e)$. Then, the constraints of updating vertex positions are:

$$\mathbf{y}(e+1) = g(\mathbf{yj}(e+1)), \tag{15}$$

where $g(\cdot)$ denotes the mapping function that accepts the initial landmark positions as input. As these landmarks undergo translation or rotation, the associated mesh vertices are translated non-linearly according to an influence strength – a tunable parameter within Autodesk Maya. The functions output the updated position of all the vertices. Note that, to our knowledge, $g(\cdot)$ is not available and is determined by Maya software.

We let the initial state of the control system be $\bar{\mathbf{x}}$, which is the rest pose. Let all the elements in the initial translation vector $\mathbf{B}$ and the initial rotational vector $\mathbf{D}$ be 0. Let the initial landmark positions $\mathbf{yj}$ be the same positions as the vertices that are selected as the corresponding landmarks at the rest pose.

3.5 Gradient descent

To cause the landmarks to move to the target pose, we use the gradient descent method:

$$\Delta\boldsymbol{\theta}(e+1) = \Delta\boldsymbol{\theta}(e) + \lambda\nabla(e)\mathbf{k}_i(e), \tag{16}$$

where $\mathbf{k}_i(e) = [0,\ldots,0,1,0,\ldots,0]^T$ is a one-shot vector with a 1 at the i-th position (the rest of the elements are all 0). $\nabla(e)$ is the single gradient (scalar value) obtained by the method in epoch e. λ is the learning rate. The gradient will be set to 0 if the reward difference between the forward and backward action is below 1×10^{-4}. The process of estimating the gradient $\Delta\boldsymbol{\theta}(e)$ by the central difference method in the finite difference method can be seen in Section B.

Moreover, to avoid the complex multiple landmarks moving at the same time and canceling each other, and to ensure the monotonical decrease of the distance after one action, we employ the coordinate descent method: We move a landmark along one axis at a time (in an epoch). We move according to the ranked magnitude of all the gradients, and we include 20% probability to choose randomly. We set up a rule that the landmarks that were moved cannot be moved again within a certain number of steps (by default, the number is the number of landmarks in a tissue).

Algorithm 1 provides the detailed steps of the algorithm pipeline for deforming a rest pose to a target pose. Note: In the algorithm chart, the "[K]" mark indicates that the step uses a kinematically based method. And the "[K+P]" mark indicates a combination of kinematically based and physically based methods.

Algorithm 1 Pipeline for deforming muscle rest pose to target pose

Input: Rest pose vertex positions $\mathbf{y}(0)$, Target pose vertex positions $\mathbf{x}$.
Output: Animation of muscle and bones.

Sample landmarks. [K]
Bind landmarks to the surrounding 20 vertices. [K]
while $d(e) \geq \xi d_0$ **do**
    Calculate the gradient $\nabla(e)$ by Eq. 39. [P]
    Update deformation parameters $\Delta\boldsymbol{\theta}(e)$ by Eq. 16.
    Apply the updated deformation parameters $\Delta\boldsymbol{\theta}(e)$ to the face model, and the movement of landmarks drives the vertices to move via Eq. 15, with the physical simulator in effect. [K+P]
    Retrieve the updated current vertex positions $\mathbf{y}(e)$ from the simulator.
    Calculate the current distance $d(e)$.
    $e \leftarrow e+1$
end while
To obtain the landmark positions $\mathbf{yj}$ of the deformed pose $\mathbf{y}$ and the intermediate positions: For $e = 1$ to *end*, incrementally apply $\Delta\boldsymbol{\theta}(e)$ to the landmark positions $\mathbf{yj}(0)$ of the rest pose $\mathbf{y}(0)$ via Eq. 14, with the physical simulator in effect. [K+P]
Animation ← The movement of landmarks drives the vertices to move via Eq. 15, with the physical simulator in effect. [K+P]

where ξ is the threshold percentage of a certain tissue, which means we expect the Euclidean distance to be smaller than ξd_0 before we claim the optimization converges. Note that the distance $d(e)$ can be generalized to each of the tissues involved in the deformation, and each tissue may require a different threshold percentage. If there is more than one tissue involved in the deformation, the while loop quits only when the distance of each of the tissues is below the corresponding threshold.

3.6 Skin registration

While the previous sections (Sections 3.1 to 3.5) detailed the generation of the internal physical motor controllers, maintaining intuitive

artistic control requires that these biomechanics smoothly drive a finalized, render-ready exterior. To bridge this gap, we propose a physically driven registration approach that tightly couples a generic, artist-provided skin mesh to the underlying tissues. By simulating this process as cloth draping over a complex collision volume, we allow the physics solver to automatically conform the exterior geometry to the personalized internal anatomy.

Because this formulation must adapt a generalized asset to subject-specific internal physics, the algorithm requires a standardized geometric starting point. Therefore, starting with a generic template skin surface, we duplicate it into two identical instances: a target skin mesh and a source (unregistered) skin mesh. The target mesh undergoes an initial rigid affine transformation—specifically, uniform scaling and translation—to roughly align its anterior contours with the superficial boundary of the internal tissues, as illustrated in Fig. 4. Conversely, the source skin mesh is uniformly inflated, akin to a balloon, ensuring it fully encapsulates the entire internal tissue volume. Moreover, we employ the lattice deformer tool in Maya to align the nose and lip with the musculature, which is important for the process of the registration.

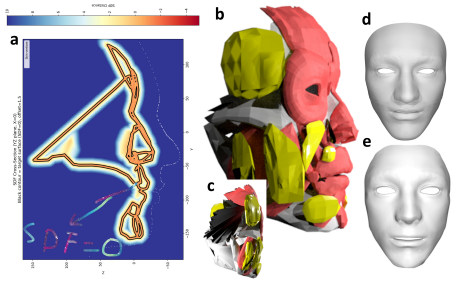

Fig. 4. The signed distance function (SDF) and the results after registration. a. The proxy collision surface by the SDF. b. The reduced musculature proxy mesh. c. The back view of the proxy mesh. d. The skin mesh before registration. e. The skin mesh after registration.

With the initial boundary conditions set, we initiate a physically based cloth simulation to drive the deformation of the registration process. Because both meshes share an identical topological structure and UV parametrization, we can establish direct correspondences. For each vertex on the inflated source mesh, we cast an attraction ray toward its corresponding vertex on the target mesh, applying continuous force until the skin accurately conforms to the underlying biomechanical structures. The convergence criterion for the registration process is evaluated per vertex; the positional update halts once a vertex approaches the target surface within a user-defined distance threshold.

Furthermore, to robustly resolve collisions between the dynamic skin mesh and the highly complex underlying anatomy, we compute a Signed Distance Function (SDF) over the internal tissues (bones, muscles, and fat pads). This SDF yields a continuous, smooth proxy collision surface, as illustrated in Fig. 4. Finally, to maximize anatomical fidelity and accurately capture the natural morphological landscape of the face, we incorporate volumetric representations of subcutaneous fat tissues. The spatial distribution and layout of these fat pads are strictly modeled following established anatomical guidelines from the plastic surgery literature [Alghoul and Codner 2013].

3.6.1 Signed Distance Function Construction. For a given query point $\mathbf{p}$, we compute the unsigned distance to each mesh:

$$d_i = \min_{\mathbf{q}\in\mathcal{S}_i} \|\mathbf{p}-\mathbf{q}\|, \quad i = 1,\dots,N_{\text{mesh}}, \tag{17}$$

where $\mathcal{S}_i$ denotes the surface of the i-th internal mesh and $\mathbf{q}$ ranges over all vertices on that surface. To prevent sharp creases at junctions between adjacent anatomical structures we employ an *exponential smooth-minimum* (soft boundary blending). Rather than evaluating all N_{mesh} meshes, we retain only the K closest (with $K = 5$ in practice) and blend their distances:

$$d_{\min}^{\text{soft}} = d_1 - \frac{1}{k}\ln\left(\sum_{i=1}^{K}\exp\big(-k\,(d_i-d_1)\big)\right), \tag{18}$$

where $d_1 \le d_2 \le \cdots \le d_K$ are the sorted per-mesh distances and k is the blending sharpness parameter (higher k yields a sharper transition; $k\to\infty$ recovers a hard minimum). The final SDF value incorporates a user-specified *skin-thickness offset* δ, which represents the desired minimum clearance between the skin surface and the internal anatomy:

$$\phi(\mathbf{p}) = d_{\min}^{\text{soft}} - \delta. \tag{19}$$

Negative values of ϕ indicate penetration (the query point lies closer than δ to the internal surface), a zero crossing marks the target isosurface, and positive values indicate that the vertex is still exterior.

In addition to the scalar distance d_i, the closest-point query on each mesh $\mathcal{S}_i$ returns the *closest surface point*

$$\mathbf{c}_i = \arg\min_{\mathbf{q}\in\mathcal{S}_i}\|\mathbf{p}-\mathbf{q}\|, \tag{20}$$

i.e. the position on the i-th internal mesh surface that is nearest to the query point $\mathbf{p}$. Because the SDF blends contributions from multiple meshes (Eq. 18), we require a single representative surface point. We therefore define the *blended closest point* $\bar{\mathbf{c}}$ as the weighted average of the individual closest points:

$$\bar{\mathbf{c}} = \sum_{i=1}^{K} w_i\,\mathbf{c}_i, \quad w_i = \frac{\exp\big(-k\,(d_i-d_1)\big)}{\sum_{j=1}^{K}\exp\big(-k\,(d_j-d_1)\big)}, \tag{21}$$

where w_i are the same exponential soft-min weights used in Eq. 18. Intuitively, $\bar{\mathbf{c}}$ lies on or near the internal tissue boundary and is smoothly interpolated across mesh junctions, whereas each individual $\mathbf{c}_i$ belongs strictly to a single anatomical surface. The outward direction from the blended surface toward the query point is then $\hat{\mathbf{n}}_{\text{out}} = (\mathbf{p}-\bar{\mathbf{c}})/\|\mathbf{p}-\bar{\mathbf{c}}\|$. This direction serves both as the SDF gradient approximation and as the push-back vector for collision response (Section 3.6.6).

After initial computation, the per-vertex SDF values are smoothed on the mesh graph using Laplacian diffusion over P passes with strength α_{sdf}:

$$\phi_i^{(p+1)} = (1-\alpha_{\text{sdf}})\,\phi_i^{(p)} + \alpha_{\text{sdf}}\,\frac{1}{|\mathcal{N}(i)|}\sum_{j\in\mathcal{N}(i)}\phi_j^{(p)}, \tag{22}$$

where $\mathcal{N}(i)$ denotes the topological 1-ring neighbourhood of vertex i. This yields a smoother implicit collision surface that hides individual mesh edges.

3.6.2 Outline Vertex Detection. Not all source-mesh vertices lie close to internal anatomy; peripheral vertices (e.g. on the forehead or neck boundary) may be far from any internal mesh. We flag vertex i as an *outline vertex* when

$$\min_{m=1,\dots,N_{\text{mesh}}} d(\mathbf{x}_i, \mathcal{S}_m) > \tau_{\text{outline}}, \tag{23}$$

where $\mathbf{x}_i \in \mathbb{R}^3$ is the position of vertex i on the source mesh, m indexes the internal anatomical meshes, $d(\mathbf{x}_i, \mathcal{S}_m)$ is the Euclidean distance from $\mathbf{t}_i$ to the closest point on surface $\mathcal{S}_m$, and τ_{outline} is a user-specified distance threshold. Outline vertices receive a fallback attraction force computed from the SDF closest-point rather than a direct vertex-to-vertex correspondence.

3.6.3 Force Model. At every simulation step each active (non-settled) vertex i accumulates a total force

$$\mathbf{f}_i = \mathbf{f}_i^{\text{attr}} + \mathbf{f}_i^{\text{strain}} + \mathbf{f}_i^{\text{bend}} + \mathbf{f}_i^{\text{lm}}, \tag{24}$$

whose components are defined below.

Target Attraction Force. The primary driving force pulls each source vertex toward its corresponding target vertex. Let $\mathbf{x}_i \in \mathbb{R}^3$ denote the *current* position of the i-th source-mesh vertex and $\mathbf{t}_i \in \mathbb{R}^3$ its corresponding position on the target mesh. The attraction force is

$$\mathbf{f}_i^{\text{attr}} = \kappa_{\text{attr}}\ \sigma(d_i,\ p_i)\ d_i\ \hat{\mathbf{d}}_i, \tag{25}$$

where $d_i = \|\mathbf{t}_i - \mathbf{x}_i\|$ is the remaining Euclidean distance to the target, $\hat{\mathbf{d}}_i = (\mathbf{t}_i - \mathbf{x}_i)/d_i$ is the unit direction pointing from the current position toward the target, κ_{attr} is the global attraction strength, and $\sigma(d_i, p_i)$ is an adaptive speed schedule used to brake the vertices upon final approach to prevent overshoot (the exact heuristic schedule is detailed in Appendix G.1. An *overshoot guard* further detects condition reversals $\hat{\mathbf{d}}_i \cdot \hat{\mathbf{d}}_i^{(0)} < 0$ and applies a gentle corrective pull.

For outline vertices the attraction defaults to the SDF closest-point target $\bar{\mathbf{c}}_i + \delta\,\hat{\mathbf{n}}_{\text{out},i}$, with a reduced speed of 0.25–0.5.

Strain Preservation Force. To maintain surface integrity, a spring-like strain force penalises deviations from the original (pre-deformation) edge lengths:

$$\mathbf{f}_i^{\text{strain}} = \kappa_{\text{strain}} \sum_{j \in \mathcal{N}(i)} \underbrace{\left(\|\mathbf{x}_j - \mathbf{x}_i\| - s\, l_{ij}^{(0)}\right)}_{\text{strain}}\ \hat{\mathbf{e}}_{ij}, \tag{26}$$

where $l_{ij}^{(0)}$ is the original edge length, $\hat{\mathbf{e}}_{ij}$ is the current edge unit vector from $\mathbf{x}_i$ toward $\mathbf{x}_j$, s is a shrink factor allowing moderate compression ($s = 0.95$), and κ_{strain} is the strain stiffness.

Bending Resistance Force. A discrete Laplacian bending term discourages local buckling:

$$\mathbf{f}_i^{\text{bend}} = \kappa_{\text{bend}} \left(\frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{x}_j - \mathbf{x}_i \right), \tag{27}$$

where κ_{bend} is the bending stiffness.

Landmark Constraint Force. Anatomically critical vertices (e.g., around the mouth, nose, and eyes) receive additional landmark constraints that anchor skin vertices to known positions on underlying muscles or bones. For a core landmark vertex i, the force is:

$$\mathbf{f}_i^{\text{lm}} = \kappa_{\text{lm}}\ \min(d_i^{\text{lm}},\ d_{\max}^{\text{lm}})\ \hat{\mathbf{d}}_i^{\text{lm}}, \tag{28}$$

where $d_i^{\text{lm}} = \|\mathbf{l}_i - \mathbf{x}_i\|$ is the Euclidean distance from the current vertex position to its landmark target $\mathbf{l}_i$ (a fixed point on the muscle or bone surface), $\hat{\mathbf{d}}_i^{\text{lm}} = (\mathbf{l}_i - \mathbf{x}_i)/d_i^{\text{lm}}$ is the corresponding unit direction, κ_{lm} is the landmark strength, and $d_{\max}^{\text{lm}}$ caps the maximum force magnitude to prevent sudden jumps.

To create smooth spatial deformations rather than sharp pin-point pulls, vertices in the topological neighbourhood of a landmark (within an influence radius r_{lm}) receive a spatially attenuated version of this force. The attenuation is defined by a cosine fall-off kernel:

$$w_{\text{inf}}(d) = \frac{1}{2}\left(1 + \cos\frac{\pi\, d}{r_{\text{lm}}}\right) \cdot 0.4, \quad d < r_{\text{lm}}, \tag{29}$$

where $d = \|\mathbf{x}_j - \mathbf{x}_i\|$ is the Euclidean distance from the neighbour vertex j to the core landmark vertex i.

This calculated influence weight $w_{\text{inf}}(d)$ operates as a direct scalar multiplier on the original force. Consequently, an influenced neighbour vertex j is pulled toward the core landmark's destination $\mathbf{l}_i$ subject to the throttled force:

$$\mathbf{f}_j^{\text{lm}} = w_{\text{inf}}(d) \cdot \kappa_{\text{lm}}\ \min(d_j^{\text{lm}},\ d_{\max}^{\text{lm}})\ \hat{\mathbf{d}}_j^{\text{lm}}. \tag{30}$$

This establishes a feathered deformation gradient; vertices immediately adjacent to anatomical landmarks are heavily guided by the underlying muscle constraint, while the pull gently diminishes to zero at the boundary of the influence radius.

3.6.4 Per-Vertex Weight Map. A per-vertex scalar weight $\omega_i \in [0, 1]$ modulates the total force on sensitive regions (lips, nose, periorbital area). The effective force becomes

$$\tilde{\mathbf{f}}_i = \omega_i\, \mathbf{f}_i. \tag{31}$$

$\omega_i = 1$ corresponds to full attraction; $\omega_i = 0.2$ is used for delicate feature areas. To avoid abrupt transitions the weight map is *feathered*: a topological BFS expansion of R rings around the core region linearly blends weights toward 1:

$$\omega_i^{(r)} = \omega_{\text{nbr,min}} + \left(1 - \omega_{\text{nbr,min}}\right)\frac{r}{R+1}, \quad r = 1, \dots, R, \tag{32}$$

where $\omega_{\text{nbr,min}}$ is the minimum weight among neighbours of vertex i.

3.6.5 Time Integration. The system state is advanced using a damped semi-implicit Euler scheme. To ensure numerical stability under complex collision scenarios, we employ strict velocity braking, adaptive step clamping, and positional overshoot guards. The full details of these numerical safeguards are provided in Appendix G.1–G.3.

3.6.6 Collision Response. After the force-based position update, a *hard collision constraint* is enforced. For each vertex we re-evaluate the raw SDF (without the offset):

$$\phi_i^{\text{raw}} = d_{\min}^{\text{soft}}(\mathbf{x}_i). \tag{33}$$

Algorithm 2 Target-Based Shrink-Wrap Registration

Require: Source mesh $\mathcal{M}_{\text{src}}$ (inflated), target mesh $\mathcal{M}_{\text{tgt}}$ (scaled down), internal tissue meshes $\{\mathcal{S}_m\}$
Ensure: Registered source mesh conforming to internal anatomy

```
Compute SDF cache for collision detection (Eqs. 17–22)
Detect outline vertices (Eq. 23)
Initialise velocities v_i ← 0, frozen set F ← ∅
for n = 1 to N_max do                          ▷ N_max: maximum iterations
    for each vertex i ∉ F do
        Compute forces: f_i^attr, f_i^strain, f_i^bend, f_i^lm   ▷ (Eqs. 25–28)
        Apply weight: f̃_i = ω_i (f_i^attr + f_i^strain + f_i^bend + f_i^lm)
        Update velocity with damping, braking   ▷ (Eqs. 41–42)
        Update position. Do clamping            ▷ (Eq.43)
    end for
    Enforce hard collision constraints          ▷ (Eq. 34)
    Local smoothing on collision-pushed vertices
    Enforce edge stretch clamp                  ▷ (Eq. 44)
    if n mod M = 0 then Global Laplacian smooth ▷ (Eq. 45)
    end if
    if n mod C_refresh = 0 then Refresh SDF cache from current positions
    end if                                      ▷ C_refresh: SDF refresh interval
    Update vertex settling / freezing status
    if convergence criterion met (Sec. G.4) then break
    end if
end for
```

If $\phi_i^{\text{raw}} < d_{\text{col}}$ (the minimum permissible distance to any internal surface), the vertex is projected outward to the safe position

$$\mathbf{x}_i \leftarrow \bar{\mathbf{c}}_i + d_{\text{col}}\,\hat{\mathbf{n}}_{\text{out},i}, \tag{34}$$

and its velocity is reset to zero. To prevent the collision correction from generating stretched triangles, a local Laplacian smoothing pass (blend factor 0.2) is applied to the pushed vertex and its 1-ring neighbourhood immediately after projection.

A *collision watchlist* accelerates the computation: a full sweep over all vertices is performed on even iterations, while odd iterations only re-check previously colliding vertices and their neighbours. Additional post-step regularisation techniques (e.g., edge stretch clamping and Laplacian smoothing passes) and the dual-convergence stopping criteria are implemented as routine stability measures, detailed in Appendix G.4.

3.6.7 Algorithm Summary. Algorithm 2 summarises the complete simulation loop, tying the force formulation and constraints together.

3.7 Muscle and bones drive skin

3.7.1 Skin attachment. To effectively transfer dynamic forces from the underlying musculature to the epidermal layer, we establish a network of spring-like constraints connecting the internal tissues to the skin mesh. This is practically implemented utilizing the attachment constraints within Ziva Dynamics. In anatomical regions characterized by significant tissue sliding, such as the buccal (cheek) area, we approximate the sliding effect through sparsely distributed fixed attachments rather than relying on Ziva's native sliding constraints. Empirical testing revealed that the native sliding solver frequently induced topological tearing and heavily damped, unnatural kinematics within the context of our high-resolution facial model.

The spatial distribution of these attachments, as shown in Fig. 5, is strictly governed by established anatomical literature [Alghoul and Codner 2013]. Specifically, strong kinematic coupling is enforced precisely at the anatomical locations of retaining ligaments, which naturally serve to rigidly tether the skin to the deeper tissues.

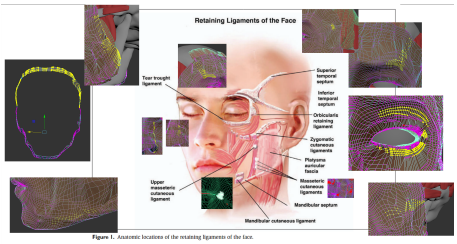

Fig. 5. The attachment positions of the skin mesh.

Conversely, the extreme structural complexity of the perioral (lip) region necessitates a degree of geometric abstraction. While our general musculoskeletal model maintains a strict separation of distinct tissue layers, the highly interwoven nature of the thin *orbicularis oris* muscle, subcutaneous fat, and epidermis in the lips is abstracted into a single, cohesive volumetric mesh. Consequently, rather than utilizing strict anatomical ligament placement, attachment points in the perioral region are assigned heuristically. This heuristic placement strategy accounts for gravitational resting states and aligns with the directional layout of the underlying *orbicularis oris* muscle fibers. Finally, to ensure reproducibility and facilitate the broader adoption of our simulation pipeline, the spatial querying (by saving the index of the attachment vertices in question) and the generation of these attachment constraints are fully automated within our provided source code.

3.7.2 Wire deformer. Relying solely on proximity-based attachments within the Ziva Dynamics framework introduces unintended coupling artifacts in the highly articulated perioral (lip) region. Specifically, the automated attachment assignment algorithm (based on the closest point method) frequently and erroneously couples the lower epidermal skin mesh to the superior *orbicularis oris* (upper lip muscle), leading to anatomically incorrect biomechanical constraints. To circumvent this algorithmic limitation, we introduce an auxiliary wire-based deformation step into our pipeline.

Furthermore, as discussed in Section 3.7.1, our volumetric abstraction of the lip tissues implies that standard elastic attachments are insufficient to independently resolve the nuanced, high-frequency deformations characteristic of human lips. To address this, we extract the boundary contour of the lip musculature (Fig. 6**a**) to serve as the driving NURBS (Non-Uniform Rational B-Spline) curves (Fig. 6**b**) for a Maya Wire Deformer. Specifically, we utilize two distinct degree-3 NURBS curves—an upper lip curve comprising 64 control vertices

and a lower lip curve comprising 45 control vertices—to ensure independent articulation of the superior and inferior labial tissues.

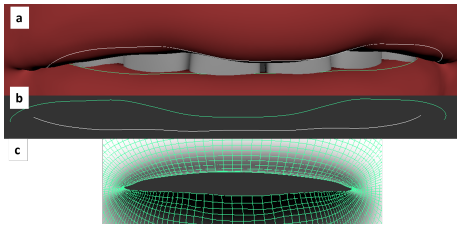

Fig. 6. The configuration of the wire deformer, illustrating the hierarchical deformation: from **a**. the muscle contour, to **b**. the wire deformer, and to **c**. the skin mesh.

In this hierarchical deformation setup, the physically simulated muscle dynamics natively drive the kinematic deformation of the extracted wire curve via the "wrap" function in Maya. The wire deformer, in turn, acts as a precise spatial guide to propagate these complex structural changes to the high-resolution skin mesh via the weighted "bind skin" function of Maya (Fig. 6**c**, where the grayscale intensity represents the influence weights of the wire deformer on the skin mesh; specifically, black denotes a full influence of 1.0, while increasing lightness indicates a non-linear falloff toward 0.0).

4 Experiments

To empirically validate our core argument—that rigorous biomechanical physics can be successfully bridged with intuitive artistic control—we implemented our proposed framework and conducted a series of expression-synthesis experiments. This section details the practical execution of our pipeline, demonstrating how theoretical physical motor controllers are translated into a functioning, render-ready facial rig. We outline the underlying geometric representations, the landmark sampling strategies, and the parameters required to achieve visually realistic animations.

4.1 Rest pose and target pose

4.1.1 Template model. A critical component of our argument is usability: to be a viable alternative to traditional kinematic rigs, our framework must seamlessly ingest standard, artist-provided geometry and automatically subject it to physical constraints. To demonstrate this capability, we acquired a high-resolution, anatomically detailed facial mesh from a commercial graphics asset market[1] to serve as our baseline. This asset, illustrated in Figure 1 in its neutral rest pose, provides the necessary distinct layers of skeletal structure and musculature required to test the robustness of our volumetric solver.

This head model contains 36 objects, including orbicularis oris, frontalis, skull, etc. The model has 241,482 polygons, and 240,124 vertices. We use Autodesk Maya software to open this head model, and then use the Ziva plugin to assign the physical attributes to the objects.

[1]https://www.cgtrader.com/3d-models/character/anatomy/head-anatomy-muscles

4.1.2 Target pose. To demonstrate the generality of our approach, we aim to use a variety of target poses from the standard facial expression coding system. The full 11 existing poses in our system are shown in Table 1. One example is the smiling to the maximum (cheese pose), as shown in Fig. 1 marked as the target pose.

To make these target poses, we assign 800 to 1000 landmarks via Poisson disk sampling [McCool and Fiume 1992] to the surface mesh, then we bind the landmarks to the mesh vertices. Finally, we manually drag the landmarks to deform the mesh to an expected standard pose. We define this manual design process as the artist's prescription. Note that we assume the positions and numbers of landmarks used in this design period are not known in the process of our control algorithm.

4.1.3 Biomechanics modeling. Three types of constraints are involved to mimic the musculoskeletal biomechanics of the face. The first is muscle attachment to bones, as shown in the red boxes in Fig. 7. The other red points are the attachment between the skin mesh and the orbicularis oris, which will be further discussed in Section 3.7. For example, in Fig. 7, the orbicularis oris is attached to the skull and the mandible, at the red points inside the pink boxes. The second type is muscle attachment to muscles, as shown in the blue boxes in Fig. 7. The "node" muscle, at the left corner of the orbicularis oris, is connected to the orbicularis oris, zygomaticus major, zygomaticus minor, etc. The third type is the fiber that is embedded in muscles, such as in the zygomaticus major, as shown in the green box in Fig. 7.

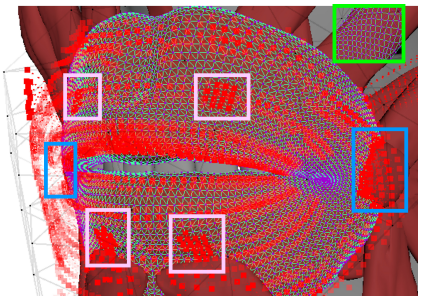

Fig. 7. Three types of constraints for muscle.

The bones in the head are treated as rigid bodies with mass. The simulation of the mandible and skull is shown in Fig. 3**b**. It is controlled by landmarks to translate and rotate.

Collision detection and constraints are enabled via the Ziva simulator. Once an action or a motion crosses the limit of Ziva's physical simulator, the simulator would stop working, and the playback would return to kinematic animation.

4.1.4 Sampling strategy. We first use the Poisson disk sampling strategy [McCool and Fiume 1992] to generate 100 landmarks as the base sample distributions. The landmark is sampled from the existing vertices of the orbicularis oris. The landmarks serve as the representatives of the whole lip muscle mesh. And the distance between any two poses is measured by the Euclidean distance of the landmarks between the two poses.

After the base landmarks, we introduce the adaptive sampling strategy to add to the base landmarks. Since the target pose and rest pose are given, we can use the Euclidean distance between each corresponding vertex of the two poses to guide the sampling process, where a vertex with a larger distance is assigned a higher probability to be chosen. We also enforce that the second wave of samples will not repeat the landmarks sampled in the Poisson disk sampling process. For orbicularis oris, we use 300 landmarks to represent it (100 from Poisson disk sampling, another 200 from adaptive sampling from target pose action unit 12 and action unit 22), and their distribution is shown in Fig. 3a.

4.2 Evaluation Metrics

For the task of approximating a rest pose to a target pose, we adopt the Euclidean distance of landmarks between the two poses. In addition, we employ screenshots of the animation and heatmap plot to visually and qualitatively evaluate the deformed face model. For energy measurement, we choose Saint Venant–Kirchhoff strain energy, elastic energy, rotational kinetic energy, and work done by the muscles.

4.3 Implementation details

We adopt the code base of the controller from one of the state-of-the-art model-free reinforcement learning (RL) algorithms: Distributional Soft Actor-Critic (DSAC) [Duan et al. 2021]. Gym, the typical reinforcement learning environment library, is employed to build our control problem. The algorithmic component is replaced by the gradient descent method. We load the Maya Python standalone module into the above code base, to control the movement of landmarks, and to retrieve the vertex positions.

The "game" (a terminology in RL, which means one pass of the whole control problem) will only be played once. During the game, the internal maximum iteration number is 100,000,000 (almost infinite in our experiments). The game will end when the distance between the current pose and the target pose is smaller than a distance threshold (for most tasks, it is 1.0% of the original distance). If multiple tissues are involved in a game, all of the distance has to reach their corresponding threshold before quitting (winning).

The number of frames of the animation is trimmed to 51 frames for the original model. Note that these particular numerical choices are experimental and can be generalized. Frames 0 to 14 are reserved for the gravity of muscles to reach almost equilibrium after the gravity takes effect at frame 0. Frame 15 to frame 50 are the samples from the whole simulation, using 1 Step 13. Moreover, the animation playback speed is 30 frames per second (FPS).

The action step length of each iteration is dependent on the differential value $\boldsymbol{\theta}$, which is 0.001.

The action step length is also capped by the biomechanical limitation of humans' motion velocity limit: we assume the maximum translational velocity of the mandible is 20 mm/s [Bourne 2002], the maximum rotational velocity of the mandible is 31.24 °/s, and the maximum velocity of facial muscle contraction is 257.6 mm/s [Ruff and Whittlesey 1991][Happak et al. 1997]. The parameters of mandible biomechanics and facial muscle movements are organized in Table 2. The detailed deduction process is given in Appendix C.

The experiments are run with the following computer environment: two Intel Core i9-9900 CPUs at 3.10 GHz, 32.0 GB RAM, equipped with Mayapy (based on Python 3.9) in Maya 2023, on Windows 11.

4.4 Results and discussions

4.4.1 The evidence of the biomechanical accuracy. To evaluate the physical fidelity and the biomechanical accuracy of our approach, we record and analyze the strain energy profiles across our comparative experiments. For the assessment of visual realism, we utilize the Euclidean distance between the simulated vertices and the target ground-truth poses as a quantitative metric. This dual-pronged evaluation demonstrates that our physically-based method achieves visual quality on par with state-of-the-art kinematic methods, while inherently preserving biomechanical integrity. Furthermore, we provide a qualitative comparison against a prominent learning-based technique [Zhao et al. 2024] via supplementary video evaluations, highlighting the differences in dynamic tissue behavior.

Initially, we evaluate our simulation by comparing our generated target pose for the Neutral-to-AA phoneme transition against an artist-sculpted ground truth (Fig. 8a and b). As demonstrated in the detailed callouts, manual sculpting techniques—such as directly pulling and dragging the mesh—can easily introduce undesirable topological distortion or "sliver" meshes. In contrast, our physically-based method preserves a uniform vertex distribution and maintains the structural integrity of the tissue (specifically, the *orbicularis oris*). Additionally, our approach naturally captures passive tissue dynamics, which are clearly visible in the deformation of the *zygomaticus major* (annotated in Fig. 8b).

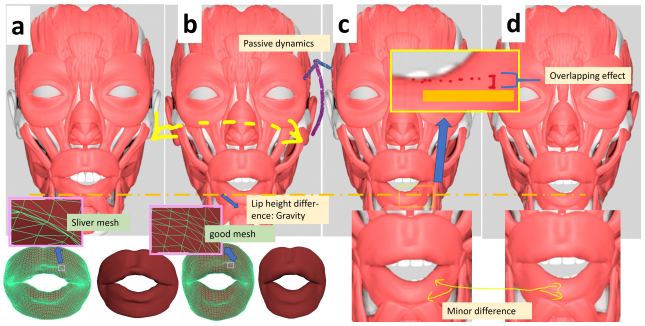

Fig. 8. Visual comparison. a. Artist's design (Ground truth). b. Our result. (Neutral to phoneme AA). c. Our result with 10% overlap (Neutral to phoneme N, to AA). d. Our result with 40% overlap (Neutral to phoneme N, to AA).

Furthermore, we analyze the biomechanical impact of overlapping sequential controllers during a chained transition (*Neutral* → *N* → *AA*). Figures 8c and d illustrate the final *AA* poses generated using a 10% and 40% temporal overlap between the transition controllers, respectively. We observe a minor morphological difference in the *orbicularis oris*: the 40% overlap yields a slightly lower position for the lower lip. We attribute this phenomenon to the timing of the mechanical forces; because the second controller (driving the

AA target) initiates earlier in the 40% overlap scenario, it encounters less accumulated mechanical resistance from the intermediate tissue deformation associated with phoneme *N*. Comparing the lower lip morphology in Figures 8a through d, a distinct vertical displacement is observed in our results (Fig. 8b–d) relative to the artist's sculpted ground truth (Fig. 8a). We attribute this lower lip position to the influence of gravitational forces, which are natively resolved by our musculoskeletal solver but omitted in the manual design. Furthermore, there is a minor difference observed in the lower lip in Fig. 8c and d. The reason is also the controller overlap effect.

Here, we detail the underlying mechanism of co-expression (co-articulation). Focusing on a single tissue—the *orbicularis oris*—we extend the experiment from Fig. 8 to simulate the spoken word "NOT" by chaining transitions from the Neutral pose through the phonemes 'N', 'AA', and 'T'. A naive, sequential animation pipeline fully completes one controller's activation before initiating the next. In contrast, to model natural biomechanical co-articulation, our approach overlaps the sequential control signals, as highlighted by the yellow box in Fig. 9a. Mechanistically, at the left boundary of this overlapping window, the active controller begins a smooth decay (ease-out), while the subsequent controller simultaneously initiates its activation (ease-in), completing the handover at the right boundary.

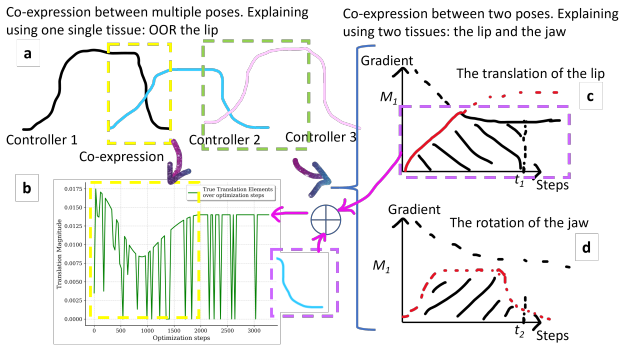

Fig. 9. Co-expression mechanics. a. The temporal overlap and cooperation between sequential controllers. b. The translation magnitude for the *orbicularis oris* over optimization steps. c. The control process for the lip translation. d. The control process for the jaw rotation.

The empirical results of this overlapping transition (from phoneme 'N' to 'AA' in the same sequential transition experiment) on the *orbicularis oris* are plotted in Fig. 9b, which illustrates the translation magnitude across optimization steps. Within the co-expression window (yellow box), the overall translation magnitude forms a smooth parabolic envelope, indicating a fluid kinematic transition. However, the graph also exhibits high-frequency oscillations dropping sharply to zero. These drops occur when the computed spatial gradient at specific landmarks becomes negligible. Because the underlying Maya simulation platform enforces a strict minimum translation tolerance, any gradient falling below this numerical threshold is explicitly clamped to zero by our approach, resulting in periodic, localized pauses in deformation.

To further elucidate the mathematical formulation of co-expression, we examine the simultaneous transition of two distinct anatomical structures: the lip (*orbicularis oris*) and the jaw (mandible). As illustrated by the black dashed lines in Figures 9c and d, the spatial gradient for a deforming tissue typically initiates at a high magnitude and asymptotically decays as the solver approaches the target pose.

Fundamentally, the total transformation (translation or rotation) required to reach the target pose is equal to the integral of this gradient curve over the optimization steps. To biomechanically synchronize the tissues such that they converge at the exact same time frame ($t_{\text{lip}} \approx t_{\text{jaw}}$), we introduce tissue-specific bounding (or capping) functions. Specifically, we utilize a sigmoid-based bounding function for the lip and a bell-shaped bounding function (consisting of two combined sigmoid curves) for the jaw.

At any given optimization step, the effective deformation update is clamped by taking the minimum operation between the natural gradient and the bounding function. This effective update corresponds to the shaded regions in Figures 9c and d.

Mathematically, this synchronization is achieved by preserving the total required deformation area while tuning the amplitude limit of the bounding functions. The relationship for a given tissue i can be expressed as:

$$C_i = \int_0^{t_i} \min\left(\nabla E_i(s),\ B_i(s; M_i)\right) ds \tag{35}$$

where C_i represents the total required deformation magnitude—expressed as either translation for the lip or Euler rotation for the jaw—necessary to achieve the target pose. The variable s denotes the current optimization step over the temporal domain $[0, t_i]$, where t_i is the total number of steps required for tissue i to reach full convergence. The effective update is governed by the minimum of the natural, unconstrained spatial gradient $\nabla E_i(s)$ and a parameterized bounding function $B_i(s; M_i)$, with M_i serving as the peak amplitude parameter or maximum velocity ceiling for the transition.

To achieve synchronous co-expression between the lip and the jaw, we must resolve the peak amplitude parameters M_{lip} and M_{jaw} subject to both temporal and spatial constraints:

$$t_{\text{lip}} \approx t_{\text{jaw}} \quad \text{s.t.} \quad C_i = \text{constant} \tag{36}$$

Critically, the total integral area C_i remains perfectly preserved regardless of the chosen M_i. This is because the total required deformation is a rigid boundary condition defined by the target pose. For example, the constant target area C_{jaw} represents the total required Euler angle rotation, which is computed *a priori* by sampling three spatial landmarks on the mandible mesh.

While this rotational estimation for the jaw provides highly accurate completion steps, estimating the expected completion steps for the lip's translation exhibits higher variance. To resolve this, we utilize an empirical estimation for the lip by multiplying the initial Euclidean distance (between the rest pose and the target pose) by a heuristically calibrated constant. Furthermore, to robustly account for any residual timing discrepancies and guarantee simultaneous convergence, we introduce an adaptive kinematic clamping mechanism for the jaw. Once the jaw's rotational progress exceeds 85%

of its total required movement, its maximum allowable rotational speed is forcibly clamped to 50% of its peak value. This programmatic deceleration allows the jaw's rotation to dynamically wait for the lip's translation, ensuring they arrive at the target state concurrently.

To systematically search for the optimal amplitude parameters M_i that satisfy these synchronization constraints, we frame this as a bi-level (second-layer) optimization problem. We employ the L-BFGS-B (Limited-memory Broyden–Fletcher–Goldfarb–Shanno with Box constraints) solver to iteratively find the optimal M_i bounds. By utilizing this solver to tune M_i, we effectively stretch or squash the temporal distribution of the force, allowing the tissues to synchronize without altering their final structural poses.

To explicitly link the subfigures: the dashed purple bounding box in Fig. 9c isolates the control envelope (the red solid line transitioning into the black solid line) corresponding exclusively to the third sequential controller (represented by the pink curve in Fig. 9a). When the kinematic influence of this isolated phase is superimposed with the preceding control signal (the blue curve), the aggregated deformation directly yields the final translation magnitude trajectory, depicted as the green curve in Fig. 9b.

To evaluate the efficacy of our overlapping controller framework in a real-world application, we synthesize a facial animation based on a reference video clip from [Zhao et al. 2024], where the character speaks the phrase "Not even you". Using the natural language sentence as input, we map the text to the phonetic sequence "N AA T IY V N Y UW" following the conventions established by [Waters and Levergood 1993]. We then design the corresponding target facial visemes. By applying our proposed physically-based approach with a preset overlap rate of 10%, we compute the intermediate landmark trajectories for each viseme transition. The resulting sequence generates a highly expressive lip-sync animation while maintaining strict biomechanical accuracy.

To provide deeper insights into the underlying solver dynamics, Figure 10 illustrates six representative simulation metrics recorded during this sequence. Specifically, Figure 10a plots the Euclidean distance between the currently simulated landmarks and the target pose during the transition from phoneme 'N' to 'Y'. Notably, the positional error temporarily increases during the first 1,000 optimization steps. This initial divergence occurs because the preceding controller (driving the 'N' pose) still exerts a dominant kinematic influence, while the subsequent controller (driving the 'Y' pose) is merely beginning its ease-in phase. Following this handover period, the distance curve decays asymptotically, ultimately converging below our preset termination threshold (defined as 10% of the initial spatial distance).

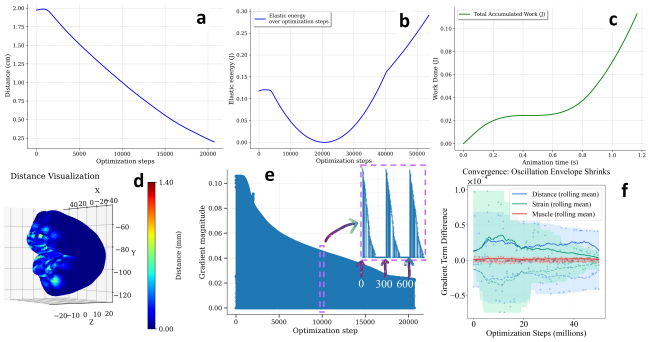

Fig. 10. Simulation dynamics and observations. (a) The landmark Euclidean distance between the current pose and the target pose (phoneme 'N' to 'Y'). (b) The elastic strain energy variation (phoneme 'V' to 'N'). (c) The accumulated muscle work (phoneme 'V' to 'N'). (d) An error heatmap comparing our simulated *orbicularis oris* mesh against the artist ground truth. (e) The gradient magnitude of the *orbicularis oris*. (f) The evolution of gradient term differences for the distance, elastic strain energy, and muscle work components (phoneme 'N' to 'Y').

We highlight another distinct transition: from phoneme 'V' to 'N'. This specific transition is notable because it requires the jaw to traverse from a raised position (above the neutral rest state) to a lowered position (below the neutral state). As depicted in Figure 10b, during the initial phase of the transition, the residual influence of the preceding controller briefly drives the jaw further away from the neutral pose, resulting in a transient increase in elastic energy. It should be noted that the energy metric plotted here serves as a computationally efficient proxy for the true elastic strain energy. Specifically, we estimate this value by tracking the arc-length displacement at the tip of the mandible using a localized linear spring approximation.

Once the active controller asserts dominance, the jaw rotates downward, passing directly through the neutral pose before settling into its lowered target position. Because elastic strain energy theoretically minimizes at the neutral rest state, this sweeping biomechanical trajectory naturally produces the pronounced parabolic energy curve observed in the plot. Furthermore, at approximately 40, 000 optimization steps, the jaw's rotation reaches the 85% completion threshold. Exactly at this juncture, our adaptive clamping mechanism is triggered, forcibly halving the maximum rotational amplitude. This algorithmic deceleration allows the jaw to dynamically wait for the slower translational convergence of the *orbicularis oris*, ensuring both tissues arrive at the final 'N' pose concurrently.

Focusing again on the phoneme 'V' to 'N' transition, we estimate the total mechanical work exerted by the muscle by integrating the work rate (mechanical power) over the deformation period. In this biomechanical context, the total work comprises both the rotational kinetic energy of the jaw and the elastic strain energy resulting from tissue deformation. The simulation spans our standard temporal window of 1.167 seconds (35 frames rendered at 30 FPS).

As illustrated in Figure 10c, the trajectory of the total accumulated work closely mirrors the elastic strain energy profile shown in Figure 10b. Notably, the accumulated work curve reaches a distinct plateau at approximately 0.5 seconds. This plateau signifies that the

mechanical power momentarily approaches zero, indicating that the muscle expends minimal energy as the jaw passes freely through its neutral, un-strained resting position. Subsequently, as the jaw continues its downward trajectory past the neutral pose toward the final 'N' target, the accumulated work curve steepens once more. This reflects the increased muscular exertion required to overcome the rising elastic resistance of the facial tissues and complete the downward deformation.

The heatmap visualization in Figure 10d demonstrates that the majority of the simulated *orbicularis oris* mesh exhibits minimal spatial deviation from the ground-truth sculpt, verifying our method's capacity to accurately approximate the target pose. The spatial distribution of this residual error is highly localized and logically consistent; the exterior, highly visible surface remains overwhelmingly within the lowest error bounds (indicated by the blue regions). The highest spatial deviations (represented by the red regions on the color scale, though fully occluded in this exterior view) occur strictly within the interior oral cavity. This localized interior error is an expected artifact of landmark under-representation, as these visually concealed regions are intentionally under-sampled during the optimization process to prioritize computational efficiency and the geometric fidelity of the visible exterior tissues.

Figure 10e plots the raw gradient magnitude of the *orbicularis oris* across the optimization steps. The initial 1,000 steps distinctly capture the anticipatory (ease-in) and carryover (ease-out) dynamics of the overlapping controllers. It is important to note that this plot visualizes the *unconstrained* raw gradient (corresponding to the dashed curve in Figure 9c). During the simulation, this raw signal is processed through the minimum bounding operation defined in Equation 35, yielding the clamped effective update (the solid curve in Figure 9c) that directly drives the landmark translations.

The zoomed-in inset, sliced at approximately step 10,000, reveals a high-frequency, sawtooth oscillation in the gradient magnitude. This looping pattern is a direct artifact of our landmark scheduling algorithm. To ensure uniform tissue deformation, the solver strictly prevents repetitive updates to the same landmark until the entire set of landmarks has been traversed. Within each traversal cycle, the algorithm greedily selects the landmark with the largest remaining gradient magnitude. Consequently, the magnitude inherently spikes at the beginning of a new cycle and drops as the cycle progresses. This also corresponds to the high-frequency oscillations in Fig. 9b with the same reason.

Furthermore, the macroscopic decreasing trend of the gradient noticeably decelerates at approximately step 14,500. This inflection point occurs precisely when the jaw's rotational progress reaches the 85% threshold. As detailed in our energy observations (Appendix Fig. Ia and b), the jaw's angular velocity amplitude is forcibly halved at this juncture. Because jaw kinematics strongly dictate the boundary conditions of the *orbicularis oris*, this sudden reduction in jaw velocity directly slows the rate of gradient descent in the lip tissue. A secondary deceleration occurs near step 18,000 when the jaw completely finishes its rotation, leaving only the residual, localized lip translation to minimize the remaining error.

Finally, in certain experimental sequences, we observe a slight rebound in the gradient magnitude during the final 10% of the optimization phase. We attribute this phenomenon to the spatial discretization limits of our landmark-based approximation. As the mesh approaches a tight local minimum, the sparse landmarks begin to exhibit conflicting gradients—effectively moving in minor opposition to cancel out residual distance errors—which manifests as a competitive, numerical increase in the overall gradient magnitude before the solver terminates.

Fig. 10f visualizes the oscillation envelope of the individual gradient term contributions, computed via finite differences (forward and backward numerical perturbations) for representative landmarks within the tissue. Our analysis reveals that the distance term consistently dictates the primary gradient direction, closely followed by the elastic strain energy term. Notably, despite sharing near-equivalent scalar weights in the objective function ($w_1 = 0.340$, $w_2 = 0.330$, $w_3 = 0.330$), the total muscle work term provides only minor, localized adjustments to the overall gradient magnitude. Crucially, the constituent gradient vectors from these three terms align cooperatively rather than antagonistically. This constructive alignment prevents severe numerical oscillation during the update iterations and ensures the kinematic fluidity of the resulting simulation. Furthermore, as the optimization converges and the oscillation envelopes shrink, the rolling mean magnitude of the distance term decreases at a slower rate than that of the strain energy. This indicates that the distance term contributes more aggressively to the convergence in the later stages. This behavior directly corresponds to the formulation in Eq. 6, where the distance penalty is a first-order (linear) term, whereas the elastic strain energy is a second-order (quadratic) term that rapidly diminishes as the tissue approaches equilibrium.

Furthermore, to account for the vast numerical scale difference between the horizontal axes of the two plots, we must dissect the solver's underlying computational search space. A massive discrepancy exists between the sheer volume of internal gradient evaluations required per search (captured in Figure 10f) and the singular, localized update steps that are actually executed after bounding function $B_i(s; M_i)$ in Eq. 35 (captured in Figure 10e).

Specifically, while Figure 10e isolates the applied gradient magnitude updates solely for the *orbicularis oris* (OOR), Figure 10f records the exhaustive internal search process across a strict sequence of all anatomical structures: the mandible (jaw), OOR, left and right *orbicularis oculi* (OOC), and the *frontalis* (uniformly sampling 20 points from this massive search for visualization). Because a single kinematic action—particularly a jaw rotation—propagates biomechanical deformation globally across multiple tissues, the solver must continuously re-evaluate the entire vertex landscape. Within this sequential update pipeline, a full search for the OOR alone requires computing finite differences for 300 spatial landmarks across all three independent Cartesian axes (x, y, z), totaling 900 scalar gradient evaluations per single update step. The algorithm sorts these 900 values, enforces the non-repeating traversal constraint, and returns the single greatest gradient for execution. It is this dense, global internal evaluation loop that directly yields the expansive 10^7 step scale observed in Figure 10f.

4.4.2 The evidence of the intuitive artistic control. Fig. 11 demonstrates our interactive, "piano-style" user interface alongside a generated simulation sequence for the phrase "Not even you". As illustrated in Figure 11a, the Maya-based interface allows users to trigger phoneme transitions sequentially via keyboard inputs, driving the underlying musculoskeletal model (Figure 11c) to deform interactively to the target poses. For temporal context, Figure 11d provides the reference audio waveform recorded by the authors, while Figure 11e maps the corresponding sequence of phoneme transitions and their designated temporal boundaries.

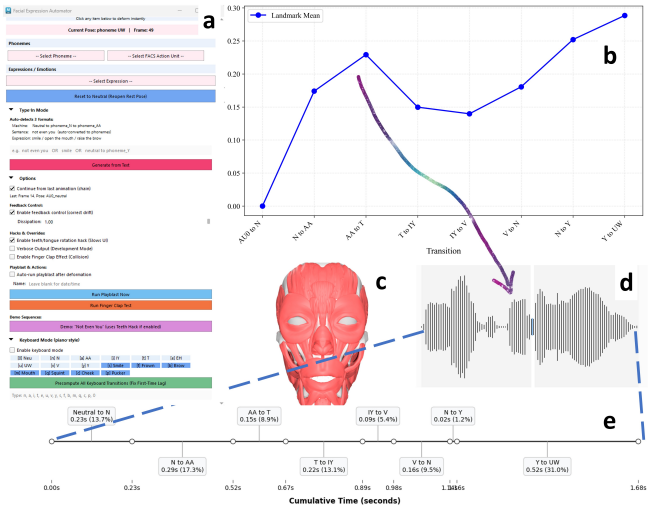

Fig. 11. The piano-style user interface and an example sequence of the sentence "Not even you". a. The Maya piano-style user interface. b. The observation of the distance mean of landmarks between the ground truth and the current pose. c. The interactive facial musculature model mesh. d. The audio waveform of this sentence recorded by the authors. e. The sequence of phoneme transitions and the time duration setting for our deformation.

A critical challenge in continuous, chained animation is the accumulation of spatial drift over multiple transitions. Figure 11b plots the landmark distance error across the example sequence. By implementing a dissipating error offset, our system actively mitigates this accumulated error, ensuring the mesh strictly converges to the target poses across extended animations.

This robust error correction and interactive control rely fundamentally on our pre-computed deformation matrix, detailed in Table 3. This matrix catalogs the stochastic deformation paths between distinct Action Units (AUs) and phonemes. Each bullet (•) represents a fully pre-computed set of deformation parameters. Because the numerical optimization required to resolve the full deformation parameters for a single transition demands an hour or more of compute time, populating this matrix offline allows the system to bypass severe runtime overhead. By leveraging these pre-computed parameters, the online forward physics simulation is reduced to merely 23 seconds for a complete 35-frame animation sequence. This massive reduction in compute time achieves the interactive responsiveness required to make the piano-style interface viable for users. Furthermore, crosses (×) denote exceptionally rare or biomechanically unnatural transitions across disparate expression groups; these are intentionally excluded from the pre-computation pipeline to maximize overall system efficiency.

4.4.3 The physical plausibility and the artistic control of the skin. Building upon our core muscle solver, we developed a high-fidelity outer skin layer driven by the underlying musculoskeletal dynamics. To ensure accurate physical force propagation from the muscles to the skin surface, the epidermal mesh must tightly conform to the complex, irregular topology of the internal tissues. To achieve this, we introduce a registration method that adapts a generic template mesh (Fig. 12a) to the personalized internal tissue landscape (Fig. 12b), ultimately yielding a production-ready, textured character model (Fig. 12c).

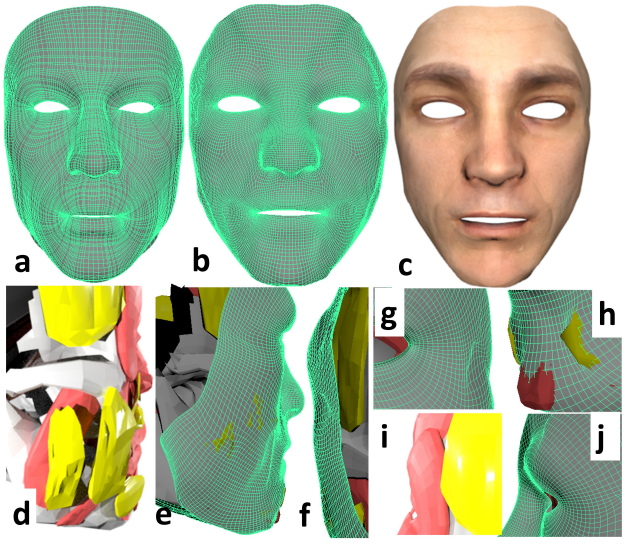

Fig. 12. Skin registration and anatomical fidelity. a. The generic template skin mesh before registration. b. The customized skin mesh after physical registration. c. The registered mesh rendered with final textures. d. and e. Right-side views demonstrating how the registered skin accurately reflects the complex landscape of the underlying internal tissues. f. and g. Detailed views of the temple region, capturing natural temporal hollowing. h. Mesh conformation around the inferior jowl compartments. i. and j. The anatomical protrusion of the inferior temporal septum, dictated by the internal tissues and accurately reflected on the registered outer surface.

Crucially, this registration process goes beyond basic boundary projection; it faithfully preserves critical, high-frequency anatomical details dictated by the underlying structures (Fig. 12d and e). As demonstrated in the detailed callouts, the resulting customized mesh naturally captures subtle biomechanical surface features, such as the characteristic temporal hollowing (Fig. 12f and g), the structural volume of the inferior jowl fat compartments (Fig. 12h), and the distinct anatomical protrusion of the inferior temporal septum (Fig. 12i and j). By strictly reflecting these underlying tissue boundaries,

the registered skin layer provides a physically sound and visually realistic foundation for the subsequent deformation.

Figure 13 highlights three representative frames extracted from the "Not even you" animation sequence. Our physical skin simulation robustly handles global passive dynamics—acting as a specialized cloth solver—by resolving gravitational forces, tensile stretch, bending resistance, and collision responses to maintain a smooth, non-intersecting manifold topology. Concurrently, a specialized wire deformer mechanism manages the highly complex, localized deformations of the *orbicularis oris* (OOR).

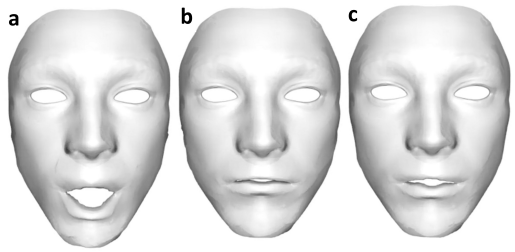

Fig. 13. Visual results of the dynamically driven skin mesh. a. The phoneme 'AA'. b. The phoneme 'V'. c. The phoneme 'UW'.

As observed in Figure 13a (phoneme 'AA'), the lower lip exhibits a clean, well-maintained parabolic arc under tension. Figure 13b captures the precise mechanical interaction during the phoneme 'V', where the lower lip actively presses and pushes upward against the upper lip. The contour in Figure 13c (phoneme 'UW') serves as a stringent deformation stress test for our model; it requires the lips to funnel and compress centrally, forming a pronounced, high-pressure pout reminiscent of Action Unit 22. Because the epidermal layer in the labial region is exceptionally thin, the final surface morphology is almost entirely dictated by the underlying OOR musculature—a biomechanical dependency that the wire deformer faithfully replicates.

Furthermore, the overall vertical extension of the lower face in Figure 13a is prominently greater than in Figures 13b and c. This macroscopic deformation is governed by the jaw, which acts as a strong kinematic boundary condition pulling downward on the chin tissue. Our outer skin layer inherently conforms to this anatomical reality, properly distributing the resulting tensile forces to create visually natural, smooth stretching across the cheek regions.

Figure 14 details the numerical observations of the skin layer during both registration and dynamic simulation. Figure 14a illustrates the convergence dynamics during the initial skin registration process. The average spatial deviation (blue curve) between the template vertices and the target mesh asymptotically approaches 11.3 Maya units (approximately 1.09 cm), representing the bulk convergence of the epidermal topology. The maximum distance metric (red curve) corresponds to regions initially furthest from the target surface, such as the forehead and chin. Conversely, the minimum distance (green curve) predominantly tracks the protruding labial (lip) regions. The initial variance across these distance metrics stems from a deliberate manual pre-alignment phase; prior to optimization, we rigidly align the eyes, nose, and lips of the template to match the vertical height of the target mesh. This spatial initialization significantly mitigates the registration difficulty across these complex, high-curvature anatomical zones.

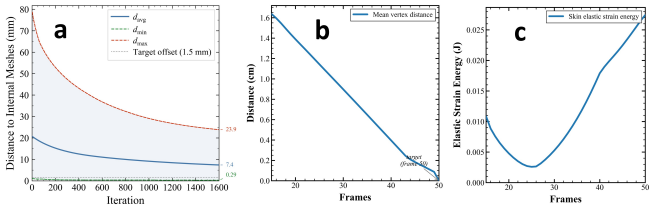

Fig. 14. Quantitative observations of the skin mesh optimization. a. Distance convergence metrics during the static skin registration process. b. The landmark Euclidean distance between the current simulated skin pose and the target skin pose during phoneme transition (phoneme 'N' to 'Y'). c. The elastic strain energy of the epidermal layer during phoneme transition (phoneme 'V' to 'N').

During the dynamic animation phase, Figure 14b tracks the Euclidean distance between the active skin vertices and their final target positions. While the macroscopic decay of this curve broadly mirrors the underlying muscular kinematics previously observed in Figure 10a, it notably lacks the transient initial increase. This deviation is a direct result of the experimental recording methodology. Unlike the internal muscle data—which is captured at the micro-resolution of individual optimization steps—the epidermal skin observations can only be extracted from the final playback frames. Because the highly co-articulated 'N' to 'Y' transition within the full sentence spans only a handful of playback frames, it lacks the temporal resolution necessary to plot a continuous dynamic curve. Therefore, to rigorously isolate and observe the skin dynamics, we executed a standard, non-overlapping transition for this specific measurement. Without the residual kinematic interference from a preceding phoneme's controller, the resulting skin distance exhibits a pure, asymptotic convergence toward zero. Because we lack an artist-sculpted ground-truth of the skin mesh, this epidermal distance metric is computed relative to our solver's final converged target state. Furthermore, Figure 14c plots the elastic strain energy of the skin layer over the course of the transition. This parabolic trajectory strongly correlates with the underlying muscle strain energy profile (recall Figure 10b), verifying that the outer epidermal layer faithfully receives and physically propagates the tensile forces exerted by the internal musculature.

4.5 Visual Fidelity Benchmarking

To evaluate the visual efficacy of our approach against state-of-the-art learning-based methods, we benchmark our simulation against the diffusion-based facial animation model from [Zhao et al. 2024]. Using the identical reference audio phrase ("Not even you"), Fig. 15 demonstrates that our physically-based model achieves an overall visual quality on par with modern data-driven and kinematic approaches.

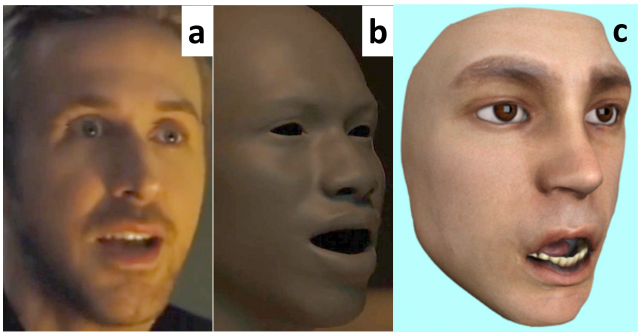

Fig. 15. Benchmark comparison with Zhao et al. [Zhao et al. 2024] using the phoneme *AA* in the phrase "Not even you". a. The reference footage from the source film. b. The baseline model pronouncing phoneme 'AA'. c. Our physically-based model pronouncing phoneme 'AA'.

Furthermore, our biomechanical solver exhibits a distinct advantage in capturing contact dynamics and soft-tissue compression. For example, during the pronunciation of the fricative consonant 'V' (Fig. 16) and the rounded vowel 'UW' (Fig. 18), the baseline diffusion model yields merely a superficial, intersecting touch between the lips. In contrast, our model naturally generates and resolves the prominent compressive force of the lower lip pressing upward against the upper lip.

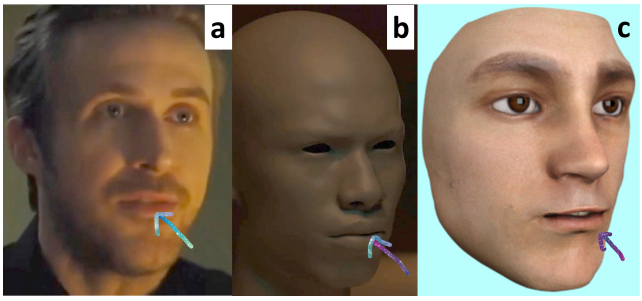

Fig. 16. Comparison of contact dynamics during phoneme 'V'. a. The reference actor from the source film. b. The baseline diffusion model [Zhao et al. 2024], showing a lack of structural compression. c. Our model, accurately capturing the compressive parabolic curve of the lower lip.

This physical accuracy is further highlighted by the detailed contours of the labial boundary. In Fig. 16a (the reference actor) and Fig. 16c (our simulation), the lower lip maintains a distinct, tension-driven parabolic curve. Conversely, this physical curvature is unresolved in the baseline output (Fig. 16b). Similarly, for the 'UW' phoneme (Fig. 18), the baseline's lower lip boundary collapses into a flat, unnatural plateau (Fig. 18b), whereas our model strictly preserves the anatomical parabolic pouting shape dictated by the underlying *orbicularis oris* (Fig. 18c).

5 Discussion and Future Work

While our experimental results demonstrate the functional success of our physical motor controllers, the broader significance of this framework lies in its ability to overcome the persistent trade-offs inherent in existing animation pipelines. By establishing a viable bridge between biomechanical physics and intuitive artistic control, our approach provides several distinct advantages over traditional methodologies.

First, current state-of-the-art learning-based methods often struggle when facial expressions are pushed to extreme deformations. Because these models rely on interpolating surface-level training data rather than computing underlying physics, they tend to "hallucinate" geometry under stress, frequently resulting in an unnatural loss of volume and structural detail. In contrast, our method establishes a biomechanical ground truth. By explicitly resolving mechanical laws for tension, compression, and collision, our framework maintains absolute structural integrity regardless of the extremity of the target pose.

Second, traditional kinematic approaches, such as blendshapes, operate in an "outside-in" manner. Artists manipulate surface vertices, and the computer interpolates the intermediate states without any computational awareness of *why* the skin is moving. Our system fundamentally reverses this paradigm. By adopting an "inside-out" architecture, we simulate the actual causal chain of human expression—from skeletal articulation to muscular contraction and subsequent epidermal deformation. This physically grounded causality extends the utility of our model beyond entertainment, offering potential downstream applications in biomechanical studies, surgical planning, and the precise analysis of speech mechanics. We are not merely animating a surface; we are simulating a functional human head.

Third, high-end physical skin registration typically requires extensive empirical 3D scans of live actors performing dozens of expressions. Our approach completely decouples physical plausibility from real-world capture data. Because our skin registration dynamically conforms a generic template mesh directly to the volumetric landscape of the underlying musculature, it eliminates the dependency on actor scans. This significantly broadens the pipeline's flexibility, allowing studios to natively apply rigorous, realistic human biomechanics to purely fictitious, stylized, or alien characters.

Finally, we address the historical barrier to entry for musculoskeletal simulation. Traditionally, animating physically-based muscle dynamics requires immense domain expertise, tedious manual parameter tuning, and prohibitively long offline compute times. Our framework bridges the gap between high-fidelity offline physics and accessible interactive control. By abstracting the heavy numerical optimization into our composable transition matrix, we allow animators to direct a complex musculoskeletal rig interactively using a "piano-style" interface. This essentially democratizes high-end physics, providing artists with the biomechanical accuracy of a medical simulation combined with the intuitive speed of a traditional rig.

While our model achieves high visual fidelity and biomechanical accuracy, several exciting avenues remain for future exploration. First, our current simulation does not explicitly model complex inter-muscle contact mechanics (e.g., adjacent muscles actively pressing against and displacing one another). Incorporating robust volumetric collision and soft-tissue compression between individual muscle

groups would further elevate the physical realism of the internal dynamics.

Second, as noted in our analysis of the transition dynamics (Figure 10b), our solver employs an algorithmic heuristic—specifically, halving the jaw's angular velocity at the 85% completion threshold—to ensure the jaw and the *orbicularis oris* arrive at the target pose concurrently. While this adaptive clamping produces visually convincing co-articulation, we currently lack *in vivo* empirical data to confirm whether human motor control strictly mirrors this synchronized deceleration. Future work will necessitate targeted biomechanical experiments to measure human neuromuscular coordination, allowing us to validate or refine this heuristic against true physiological behavior.

Finally, our current objective function computes mechanical work based solely on the rotational kinetic energy of the jaw and the elastic strain energy of the musculature. A natural next step is to incorporate the elastic strain energy of the skin deformation directly into the global optimization loop (this may require the target pose of the skin mesh from the artists' design). Creating a fully coupled system that holistically minimizes energy across both the internal tissues and the outer epidermal layer will yield an even more rigorous, physically unified model.

6 Conclusion

In this paper, we have demonstrated that rigorous biomechanical physics and intuitive artistic control are not mutually exclusive. To bridge these two domains, we presented a comprehensive, inside-out facial animation framework that abstracts complex internal anatomies into intuitive physical motor controllers.

By leveraging representative spatial landmarks and a custom gradient descent solver that minimizes both geometric distance and biomechanical energy, our approach successfully drives anatomical musculature to match artist-sculpted target poses. To extend this solver to continuous animation, we introduced a composable expression transition framework. Central to this is our novel co-expression mechanism, which seamlessly blends overlapping controllers to ensure smooth, natural transitions between phonetic states. By pre-computing these deformation trajectories, we bypass heavy runtime overhead, empowering users with an interactive, "piano-style" interface capable of driving complex, sequential facial animation interactively. Finally, we developed a robust physical skin registration pipeline that tightly conforms a high-fidelity epidermal mesh to the underlying tissues. Ultimately, this framework ensures that the complex, physiologically grounded dynamics of the internal muscle solver are vividly propagated to the surface, granting artists unprecedented physical realism without sacrificing creative control.

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A The results of other expressions and phonemes

Table 1 is the selected action-unit names and descriptions.

Index	Name	Description
1	AU 0	Neutral pose
2	AU 6	Cheek raiser
3	AU 12	Smile
4	AU 22	WUU sound
5	AU 12L	Left asymmetric smile
6	AU 7	Lid tightener
7	AU 25	Lips part
8	AU 4	Brow lowerer
9	AU 1	Inner brow raiser
10	AU 2	Outer brow raiser
11	AU 6+12+25ii	Cheese (maximum)
12	Phoneme N	*n*et
13	Phoneme AA	p*o*t
14	Phoneme T	*t*est
15	Phoneme IY	b*ea*t
16	Phoneme V	*v*et
17	Phoneme Y	*y*et
18	Phoneme UW	l*u*te
19	Phoneme EH	b*e*t

Table 1. Selected action-unit names and descriptions. "ii" means extra high intensity. "L" means the expression on the left side only.

B Gradient descent equations

$$r_{fw}(e) = - \parallel \mathbf{x} - (\mathbf{y}(e) + \delta \mathbf{k}_i(e)) \parallel_F^2, \tag{37}$$

$$r_{bw}(e) = - \parallel \mathbf{x} - (\mathbf{y}(e) - \delta \mathbf{k}_i(e)) \parallel_F^2, \tag{38}$$

$$\nabla(e) = -\frac{|r_{fw}(e) - r_{bw}(e)|}{2\delta}, \tag{39}$$

where $r_{fw}(e)$ and $r_{bw}(e)$ are the reward of the forward and backward differential value to the current position, respectively. $\delta = 0.001$ is the differential value.

Note that because the mapping function $g(\cdot)$ between the control of a landmark and the translation of a vertex corresponding to this landmark, is an unknowable black box reserved in the Maya platform, we use the central difference method in the finite difference method to estimate the gradient in Eq. 39.

C Facial biomechanics

There is a limit on the translation and rotation speed of facial tissues. Regarding the rotation speed of the mandible, the average chewing speed is 20 mm/s [Bourne 2002]. The maximum speed is 130 mm/s [Carossa et al. 2020].

We use 25 mm (measured by the authors using themselves as the subject) as the external arc length of the mandible. The radius from the center of the rotation to the middle teeth is 17 cm for an adult male human[2]. Thus, we could obtain the rotation angle of the range as 8.43 degrees. We observe that people spend 0.2698 s when they open their mouths while chewing gum at a fast speed [Bourne 2002]. Then the rotational velocity is calculated as 31.24 °/s.

The facial muscle contraction velocity is 4.6 muscle lengths per second at most (excluding the extremely fast part such as eye lid blinking) [Ruff and Whittlesey 1991]. Assume the average facial muscle length is 56 mm [Happak et al. 1997]. Then the maximum velocity of facial muscle contraction is calculated as 257.6 mm/s.

Moreover, we need to convert this to the unit for the head model. We measure the mandible of the virtual head model, and compare it with the measurement of the mandible of one of the authors of this survey. In Maya, the width of the mandible is 155.92 units, and the author's counterpart is 15 cm. Thus, the unit mapping is 0.0962 cm per Maya unit.

In summary, the bottom line of motion velocity limit is: we assume the maximum translational velocity of the mandible is 20 mm/s (or 0.208 Maya unit/s), the maximum rotational velocity of the mandible is 31.24 °/s, and the maximum velocity of facial muscle contraction is 257.6 mm/s (or 2.678 Maya unit/s). The parameters of mandible biomechanics and facial muscle movements are organized in Table 2.

Parameter	Value
Maximum translational velocity of the mandible	20 mm/s
Maximum rotational velocity of the mandible	31.24 °/s
Maximum velocity of facial muscle contraction	257.6 mm/s
Unit mapping in Maya	0.0962 cm per Maya unit
External arc length of the mandible	25 mm
Radius from the center of rotation to the middle teeth	17 cm
Rotation angle range of the mandible	8.43°
Time to open mouth while chewing gum at high speed	0.2698 s

Table 2. Parameters of mandible biomechanics and facial muscle movements.

D The transition matrix

Please see Table 3.

E Variables and parameters in the objective function and Hill muscle model

Please see Table 4

F Justification for Adapted Force–Length Parameterization

In our formulation of the active force–length relationship (Equation 10), we adapt the standard biomechanical model [Zajac 1989] by substituting the absolute anatomical translation of the fibre length with $\bar{\Delta}d$, representing the average remaining distance to the target pose. This parameterization implicitly assumes that the target pose corresponds to a localized minimum of strain energy and muscle force. While this is a strong assumption—and anatomically untrue for highly deformed target poses such as the phoneme AA—it is a necessary mathematical adaptation to ensure optimization convergence, speed, and sequential stability. The justification for this adaptation is threefold:

[2]https://www.theskeletalsystem.net/skull-bones/mandible.html

Table 3. Transition Matrix between Action Units (AUs) and Phonemes based on Stochastic Deformation Paths. Verified transitions are marked with •, explicitly restricted transitions between groups with ×, and missing intra-group paths are left blank.

	Target Pose																			
Source Pose	AU0	AU6	AU12	AU22	AU12L	AU7	AU25	AU4	AU1	AU2	AU2L	AU2R	Cmb	N	AA	EH	T	IY	V	Y	UW
AU0 (Neutral)	–	•	•	•		•	•	•	•	•	•	•		•	•	•	•	•	•	•	•
AU6	•	–	•	•		•	•	•	•	•				×	×	×	×	×	×	×	×
AU12	•	•	–	•		•	•	•	•	•				×	×	×	×	×	×	×	×
AU22	•	•	•	–		•	•	•	•	•				×	×	×	×	×	×	×	×
AU12L					–									×	×	×	×	×	×	×	×
AU7	•	•	•	•		–	•	•	•					×	×	×	×	×	×	×	×
AU25	•	•	•	•			–	•	•	•				×	×	×	×	×	×	×	×
AU4	•	•	•	•			•	–	•	•				×	×	×	×	×	×	×	×
AU1	•	•	•	•			•	•	–	•				×	×	×	×	×	×	×	×
AU2	•	•	•	•			•	•	•	–				×	×	×	×	×	×	×	×
AU2L	•										–										
AU2R	•											–									
Combo†													–	×	×	×	×	×	×	×	×
Phoneme N	•	×	×	×	×	×	×	×	×	×			×	–	•	•	•	•	•	•	•
Phoneme AA	•	×	×	×	×	×	×	×	×	×			×	•	–	•	•	•	•	•	•
Phoneme EH	•	×	×	×	×	×	×	×	×	×			×	•	•	–	•	•	•	•	•
Phoneme T	•	×	×	×	×	×	×	×	×	×			×	•	•	•	–	•	•	•	•
Phoneme IY	•	×	×	×	×	×	×	×	×	×			×	•	•	•	•	–	•	•	•
Phoneme V	•	×	×	×	×	×	×	×	×	×			×	•	•	•	•	•	–	•	•
Phoneme Y	•	×	×	×	×	×	×	×	×	×			×	•	•	•	•	•	•	–	•
Phoneme UW	•	×	×	×	×	×	×	×	×	×			×	•	•	•	•	•	•	•	–

† Combination of AU6, AU12, and AU25ii (No deformation paths in current dataset).

Table 4. Variables and parameters in the objective function (Eq. 6) and Hill muscle model (Eq. 9). All lengths in the code are converted from Maya units to metres (1 Maya unit = 9.62×10^{-4} m) or centimetres where noted.

Symbol	**Description**	**Value**	**Unit**
Objective Function Parameters			
$w_1,\ w_2,\ w_3$	Weights for the distance, strain-energy, and muscle-work terms	0.34, 0.33, 0.33	—
d_{remain}	Current Euclidean distance between simulated and target vertex positions, $\|\mathbf{x}-\mathbf{y}\|_F$	—	Maya unit
d_{init}	Initial Euclidean distance between rest-pose and target vertices (per-tissue)	—	Maya unit
k	Stiffness of soft tissue (linear spring constant)	10 000	N/m
ΔL	Strain distance: deformation magnitude from current to target position	—	Maya unit
s	Unit-conversion: Maya units to metres	9.62×10^{-4}	m / Maya unit
$E_{\max}$	Maximum expected strain energy (normalization constant)	11.1	J
c	Amplification factor for the muscle-work gradient	1000	—
$W_{\max}$	Maximum expected muscle work (normalization constant)	7.32×10^{-5}	J
Δp_i	Translational displacement of the i-th landmark from rest	—	Maya unit
Hill Muscle Model Parameters			
α	Degree of muscle activation	0.8	—
$F_{\max}$	Maximum isometric force per landmark (17 N total / 300 landmarks)	0.0567	N
l_0	Anatomical muscle fibre length	2.0×10^{-2}	m
l_0	Optimal fibre length	2.0×10^{-2}	m
σ	Width of the Gaussian length–tension curve	0.4	—
$v_{\max}$	Maximum shortening velocity	∼0.235	cm/s
K	Shape factor for the force–velocity curve	0.18	—
a	Hill's constant (force–velocity curve parameter)	0.12	—
b	Hill's constant, derived as $v_{\max} \cdot a/(F_{\max} \cdot K)$	derived	—
k_{PEE}	Stiffness of the parallel elastic element	0.5	N/cm
k_{SEE}	Stiffness of the series elastic element	0.3	N/cm
l_{tendon}	Tendon stretch	0.05	cm

1. Mitigation of Conflicting Gradients. The primary goal of the optimization process is to minimize the positional error between the current state and the target pose. If true absolute strain energies and muscle forces are strictly enforced in the objective function, they act as severe damping forces. For instance, when deforming from a neutral rest pose to an extreme phoneme pose, the true bio-energy gradients resist the distance-driven gradients. This resistance frequently leads to vanishing gradients, halting the optimization

before convergence. By re-centering the energy formulation relative to the target distance, we prevent the physical constraints from fighting the primary kinematic objective.

2. Simplicity of Facial Trajectories. Unlike complex articulated limb movements (e.g., an arm navigating obstacles to grasp an object), facial landmark trajectories during expression or speech are highly constrained and predominantly linear. Because the paths of these landmarks over short distances do not require complex, energy-guided spatial exploration, enforcing a strictly accurate energy landscape does not yield significantly wiser or more natural trajectories than our results.

3. Sequential Soft Constraints and Boundary Conditions. In practical facial animation, isolated transitions (A → B) are ultimately chained into continuous sequences (A → B → C). In this context, intermediate poses (such as individual phonemes) act as transient boundary conditions. As long as the simulation reaches the "right vicinity" of these intermediate targets, the subsequent transitions remain smooth and visually natural. Therefore, imposing stringent, absolute energy minimization on every intermediate frame is overly restrictive. Our formulation acts as a "soft constraint," prioritizing spatial proximity for intermediate frames and reserving strict global energy accuracy for the final transition back to a global rest pose.

G Simulation Implementation Details

This appendix provides the precise numerical safeguard mechanisms, time-integration heuristics, and default parameter values used to ensure the unconditional stability of the primary skin-registration formulation.

G.1 Adaptive Speed Schedule

The attraction force is modulated by $\sigma(d_i, p_i)$, an adaptive speed factor that depends on both the absolute distance and the fractional *progress* $p_i = 1 - d_i/d_i^{(0)}$ (where $d_i^{(0)}$ is the initial source-to-target distance). The speed schedule implements a braking zone with conditions fulfilled with priority from top to bottom:

$$\sigma(d_i, p_i) = \begin{cases} 0.05 & p_i > 0.95 \text{ (final 5 \%)}, \\ 0.10 & p_i > 0.90, \\ 0.25 & p_i > 0.80, \\ 2.0 & d_i > 10, \\ 1.5 & d_i > 5, \\ 1.0 & d_i > 1, \\ 0.5 & d_i > 0.2, \\ 0.25 & \text{otherwise.} \end{cases} \tag{40}$$

G.2 Time Integration and Adaptive Braking

The simulation advances using a semi-implicit Euler scheme. At every iteration the velocity is updated as:

$$\mathbf{v}_i^{(n+1)} = \gamma\, \mathbf{v}_i^{(n)} + \Delta t\, \tilde{\mathbf{f}}_i^{(n)}, \tag{41}$$

where $\gamma \in (0, 1)$ is the global velocity damping coefficient, Δt is the simulation time step, and $\tilde{\mathbf{f}}_i^{(n)}$ is the weighted force from Eq. 31. The gravity of each vertex is simplified to be 1.0 unit, so it is removed from Eq.41.

To prevent violent oscillations when approaching target states, an adaptive *progress-dependent braking* coefficient $\beta(p_i) \in (0, 1]$ is applied to the velocity:

$$\mathbf{v}_i^{(n+1)} \leftarrow \beta(p_i)\, \mathbf{v}_i^{(n+1)}, \quad \beta(p_i) = \begin{cases} 0.1 & p_i > 0.95 \text{ (final 5 \%)}, \\ 0.3 & p_i > 0.90, \\ 0.5 & p_i > 0.80, \\ 1.0 & \text{otherwise}, \end{cases} \tag{42}$$

where $p_i = 1 - d_i/d_i^{(0)}$ represents the fractional approach progress. The positions are then updated via $\mathbf{x}_i^{(n+1)} = \mathbf{x}_i^{(n)} + \Delta t\, \mathbf{v}_i^{(n+1)}$.

G.3 Numerical Stability Clamps

Global Movement Clamp. Per-vertex displacements are strictly clamped to a maximum movement limit $\Delta_{\max}$ to prevent mathematical divergences during extreme deformations:

$$\mathbf{x}_i^{(n+1)} \leftarrow \mathbf{x}_i^{(n)} + \min\left(1,\ \frac{\Delta_{\max}}{\|\mathbf{x}_i^{(n+1)} - \mathbf{x}_i^{(n)}\|}\right)\left(\mathbf{x}_i^{(n+1)} - \mathbf{x}_i^{(n)}\right). \tag{43}$$

Edge Stretch Clamp. To prevent catastrophic mesh topological distortion, any edge that stretches beyond twice its original rest length is corrected immediately:

$$\text{if} \quad \|\mathbf{x}_j - \mathbf{x}_i\| > 2\, l_{ij}^{(0)} : \quad \begin{cases} \mathbf{x}_i \leftarrow \mathbf{x}_i + 0.3\, \Delta l\, \hat{\mathbf{e}}_{ij}, \\ \mathbf{x}_j \leftarrow \mathbf{x}_j - 0.3\, \Delta l\, \hat{\mathbf{e}}_{ij}, \end{cases} \tag{44}$$

where $\Delta l = \|\mathbf{x}_j - \mathbf{x}_i\| - 2\, l_{ij}^{(0)}$ and both endpoints are resiliently pulled inward symmetrically.

Laplacian Smoothing. Every M iterations, a global Laplacian smoothing pass serves as high-frequency noise regularisation:

$$\mathbf{x}_i \leftarrow (1 - \alpha_{\text{sm}})\, \mathbf{x}_i + \alpha_{\text{sm}}\, \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \mathbf{x}_j, \tag{45}$$

with smoothing strength α_{sm}.

G.4 Convergence and Settling

A simulation vertex is declared *settled* into a frozen state when its absolute positional step and remaining distance to the target fall below threshold ε_s for N_s consecutive iterations. Frozen vertices no longer evaluate physics loops, drastically reducing integration times. Furthermore, the global simulation yields early termination if the average per-vertex displacement:

$$\bar{\Delta} = \frac{1}{N_{\text{vtx}}} \sum_{i=1}^{N_{\text{vtx}}} \|\mathbf{x}_i^{(n+1)} - \mathbf{x}_i^{(n)}\| \tag{46}$$

stays below ε_c for N_c consecutive condition checks.

G.5 Simulation Parameters

Table 5 provides the verified default parameter values configured for this study.

Table 5. Variables and parameters in the objective function (Eq. 6) and Hill muscle model (Eq. 9). All lengths in the code are converted from Maya units to metres (1 Maya unit = 9.62×10^{-4} m) or centimetres where noted.

Symbol	Description	Value	Unit
Objective Function Parameters			
$w_1,\ w_2,\ w_3$	Weights for the distance, strain-energy, and muscle-work terms	0.34, 0.33, 0.33	–
d_{remain}	Current Euclidean distance between simulated and target vertex positions, $\|\mathbf{x}-\mathbf{y}\|_F$	–	Maya unit
d_{init}	Initial Euclidean distance between rest-pose and target vertices (per-tissue)	–	Maya unit
k	Stiffness of soft tissue (linear spring constant)	10 000	N/m
ΔL	Strain distance: deformation magnitude from current to target position	–	Maya unit
s	Unit-conversion: Maya units to metres	9.62×10^{-4}	m / Maya unit
$E_{\max}$	Maximum expected strain energy (normalization constant)	11.1	J
c	Amplification factor for the muscle-work gradient	1000	–
$W_{\max}$	Maximum expected muscle work (normalization constant)	7.32×10^{-5}	J
Δp_i	Translational displacement of the i-th landmark from rest	–	Maya unit
Hill Muscle Model Parameters			
α	Degree of muscle activation	0.8	–
$F_{\max}$	Maximum isometric force per landmark (17 N total / 300 landmarks)	0.0567	N
l_0	Anatomical muscle fibre length	2.0×10^{-2}	m
l_0	Optimal fibre length	2.0×10^{-2}	m
σ	Width of the Gaussian length–tension curve	0.4	–
$v_{\max}$	Maximum shortening velocity	~0.235	cm/s
K	Shape factor for the force–velocity curve	0.18	–
a	Hill's constant (force–velocity curve parameter)	0.12	–
b	Hill's constant, derived as $v_{\max} \cdot a/(F_{\max} \cdot K)$	derived	–
k_{PEE}	Stiffness of the parallel elastic element	0.5	N/cm
k_{SEE}	Stiffness of the series elastic element	0.3	N/cm
l_{tendon}	Tendon stretch	0.05	cm

H Temporal Mapping and Simulation Configurations

In this section, we detail the temporal configurations of our musculoskeletal simulation, explicitly defining the mapping between real-world phonetic timing, software frame rates, and our solver's optimization steps.

H.1 Frame Rate and Global Duration

Our simulation environment in Autodesk Maya is configured to a standard animation frame rate of 30 frames per second (FPS), yielding a temporal resolution of approximately 33.3 milliseconds (ms) per frame. A standard continuous simulation sequence in our experiments is executed over 35 frames. Therefore, the total real-world duration of the simulation block is calculated as:

$$T_{total} = 35 \text{ frames} \times \left(\frac{1}{30} \text{ seconds}\right) \approx 1.167 \text{ seconds (or 1167 ms)}. \tag{47}$$

H.2 Phonetic Transition Timing

Based on established research on phoneme transition [Giroud et al. 2024], we assume an average transition speed of 65 ms per phoneme. Using this baseline, we can calculate the active deformation time for various phonetic sequences:

- **Short Sequence (Neutral → N → AA → T):** This sequence comprises three sequential transitions. The expected duration is 3×65 ms = 195 ms.
- **Full Sentence ("Not even you"):** This sentence maps to an eight-transition phonetic sequence (Neutral → N → AA → T → IY → V → N → Y → UW). The expected active transition duration is 8×65 ms = 520 ms.

H.3 Solver Optimization Steps and Co-expression

To compute the physical deformation across the 35-frame sequence, our numerical solver is allocated around 10,000 optimization steps. By mapping these steps to the global simulation duration (1167 ms), we establish the solver's temporal resolution:

$$\Delta t_{step} = \frac{1167 \text{ ms}}{10,000 \text{ steps}} \approx 0.1 \text{ ms per optimization step}. \tag{48}$$

This high-resolution step size is critical for our co-expression mechanism. We allocate 1,000 optimization steps specifically for the anticipatory (ease-in) and carryover (ease-out) effects. Given our step resolution, this provides a strictly defined 100 ms temporal window for smooth biomechanical co-expression between adjacent phonetic states.

I The simulation observations of the energy

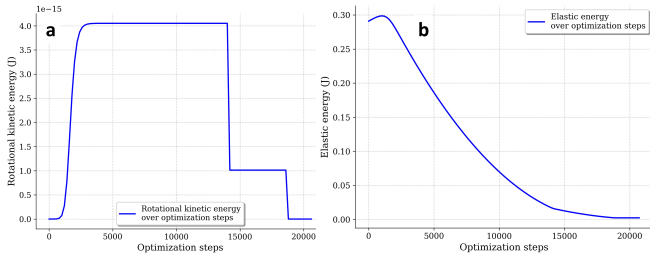

Fig. 17. The simulation observations of the energy. a. The rotational kinetic energy over optimization steps (from phoneme N to phoneme Y). b. The elastic strain energy over optimization steps (from phoneme N to phoneme Y).

J Visual fidelity benchmarking

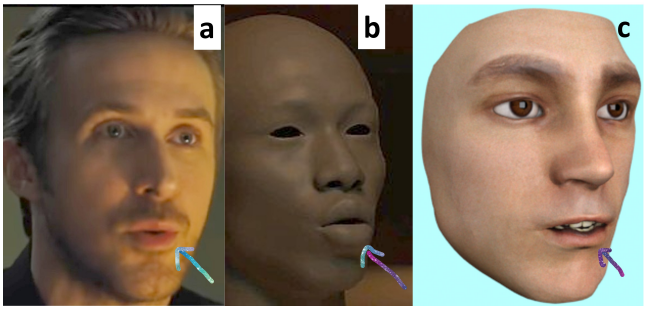

Fig. 18. Comparison of labial contours during phoneme 'UW'. a. The reference actor from the source film. b. The baseline model [Zhao et al. 2024], where the lower lip flattens into an unnatural plateau. c. Our model, preserving the structurally sound parabolic pout.

K Ablation Studies

To validate the efficacy and necessity of our core algorithmic components, we conduct two primary ablation studies. These experiments isolate specific mechanisms within our framework to demonstrate their direct impact on the physical accuracy and kinematic fluidity of the resulting facial animation.

K.1 Objective Function Components

To evaluate the individual contributions of the objective function components, we conducted five comparative ablation experiments, summarized in Table 6. Surprisingly, all five experimental configurations exhibited stable and successful convergence trajectories. One example is the gradient term difference plot of group 3 in Fig. 20, and the observation aligns with the one in Fig. 10. We attribute this robustness to the underlying mathematical formulation: all three objective terms are fundamentally coupled to the Euclidean distance between the current and target poses, albeit evaluated at different polynomial orders. Specifically, the geometric distance penalty is a first-order (linear) term, the elastic strain energy is a second-order (quadratic) formulation, and the muscle work term incorporates distance implicitly through force-length relationships. While this creates a degree of mathematical redundancy among the terms, this deliberate coupling ensures global convergence regardless of the active subset. Crucially, this modular design provides users with an intuitive, physically grounded mechanism to tune the objective weights based on their specific animation or biomechanical requirements.

Table 6. Comparative ablation study of objective function weights. A checkmark (✓) indicates the active terms.

Experiment	Dist.	Strain	Work	Description
Group 1 (Ours)	✓	✓	✓	Evenly distributed weights
Group 2	✓			Purely geometric driving
Group 3		✓	✓	Purely internal biomechanics
Group 4	✓	✓		Distance and strain only
Group 5	✓		✓	Distance and muscle work only

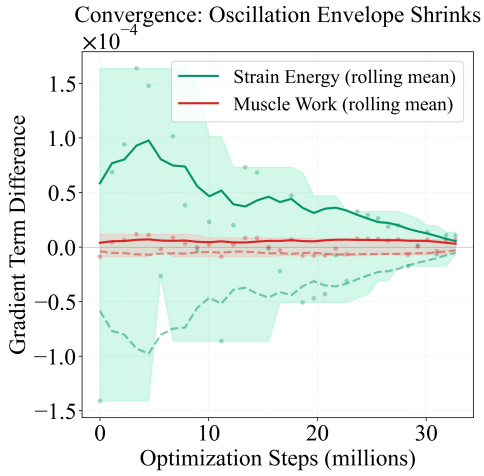

Fig. 19. The evolution of gradient term differences for the distance, elastic strain energy, and muscle work components (from the neutral pose to AU22).

K.2 Overlapping Controllers

We demonstrate the critical role of our overlapping controller mechanism in handling co-expression. In this ablation study, we disable the temporal overlap—specifically the anticipatory ease-in and carryover ease-out phases—forcing the solver to execute strictly sequential, isolated phoneme transitions. To evaluate this, we designed an experiment simulating the rapid, repeated vocalization of the syllable "va" (as in the French phrase *Ça va bien?*) ten consecutive times. Empirical self-testing established a maximum human vocalization speed of 2.58 seconds for this sequence. Achieving this rapid phonetic firing physically necessitates co-articulation; attempting to enunciate the sequence with strictly isolated, non-overlapping muscle activations at this speed exceeds human physiological limits.

We compared the resulting simulations at 0% overlap against our proposed 40% overlap, visualizing the geometric divergence via a

heatmap. As shown in Fig. 20, the 0% overlap condition suffers from an unnatural accumulation of downward momentum. Without the anticipatory tension of the subsequent phoneme, the *orbicularis oris* muscle and surrounding tissues sag progressively lower with each iteration, ultimately resulting in the unnatural occlusion of the upper teeth. Conversely, the 40% overlap condition successfully mitigates this gravity-induced drift. Because the ease-in phase of the subsequent phonetic controller activates before the jaw reaches its lowest kinematic boundary, it acts as an anticipatory muscular braking mechanism, decelerating the downward trajectory and preemptively initiating the upward pull. We conclude that this overlapping activation pattern is essential for accurately replicating the biomechanical fluidity and structural integrity of human speech.

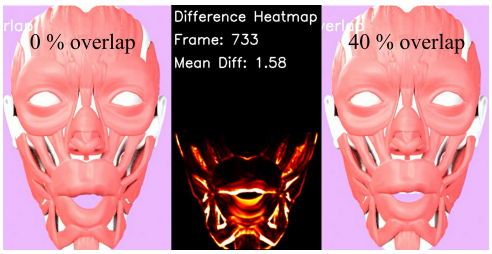

Fig. 20. Heatmap visualization of the final frame after 10 rapid, repeated simulations of the syllable "va". The comparison highlights the severe geometric divergence between strictly sequential activation (0% overlap) and co-articulated activation (40% overlap). Controller overlap prevents the unnatural, gravity-induced tissue sagging and tooth occlusion seen in the 0% condition by introducing an anticipatory muscular braking effect.